Supplementary notes
BI-2026-3: prescribed-time contracting-boundary control

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These notes provide detailed derivations, numerical evaluation methods and visual-feedback methods accompanying the manuscript. Main-text equations and figures are identified explicitly. The following data appendix contains complete task comparisons and sensitivity results.

1 Null space, compatibility, and orthogonal projectors

From main-text Eq. (29), we obtain

$$\begin{aligned} q_1 &= dLc_x, \\ q_2 &= dL\left(\tfrac{1}{2}c_x + \tfrac{\sqrt{3}}{2}c_y\right), \\ q_3 &= dL\left(-\tfrac{1}{2}c_x + \tfrac{\sqrt{3}}{2}c_y\right), \\ q_4 &= -dLc_x, \\ q_5 &= dL\left(-\tfrac{1}{2}c_x - \tfrac{\sqrt{3}}{2}c_y\right), \\ q_6 &= dL\left(\tfrac{1}{2}c_x - \tfrac{\sqrt{3}}{2}c_y\right). \end{aligned} \tag{1}$$

Consequently,

$$q_4 = -q_1, \quad q_5 = -q_2, \quad q_6 = -q_3, \quad q_1 - q_2 + q_3 = 0. \tag{2}$$

These relations characterize the two-dimensional tendon-shortening subspace compatible with the ideal pure-bending kinematics. The Gram matrix calculation is

$$\begin{aligned} \boldsymbol{C}^{\mathsf{T}}\boldsymbol{C} &= \begin{bmatrix} \sum_{i=1}^{6}\cos^2\phi_i & \sum_{i=1}^{6}\cos\phi_i\sin\phi_i \\ \sum_{i=1}^{6}\cos\phi_i\sin\phi_i & \sum_{i=1}^{6}\sin^2\phi_i \end{bmatrix} \\ &= \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = 3\boldsymbol{I}_2. \end{aligned} \tag{3}$$

Premultiplication of main-text Eq. (30) by $\boldsymbol{C}^{\mathsf{T}}$ therefore gives the exact inverse on the pure-bending subspace:

$$\boldsymbol{c} = \frac{1}{3dL}\boldsymbol{C}^{\mathsf{T}}\boldsymbol{q}. \tag{4}$$

Its two entries are

$$c_x = \frac{q_1 + \frac{1}{2}q_2 - \frac{1}{2}q_3 - q_4 - \frac{1}{2}q_5 + \frac{1}{2}q_6}{3dL}, \tag{5}$$

$$c_y = \frac{\frac{\sqrt{3}}{2}(q_2 + q_3 - q_5 - q_6)}{3dL}. \tag{6}$$

When the three opposite-pair relations hold, these reduce to

$$c_x = \frac{2q_1 + q_2 - q_3}{3dL}, \quad c_y = \frac{q_2 + q_3}{\sqrt{3}\,dL}. \tag{7}$$

The factor $1/(3dL)$ is thus analytical, not a fitted pseudoinverse. In a physical platform, unequal tendon offsets, routing angles, or transmission gains generally destroy the symmetry of (3). The resulting tendon-to-curvature map should therefore be calibrated from independent shape measurements and verified to have full column rank. Because $\operatorname{rank}(\boldsymbol{C}) = 2$, the six-dimensional actuation space decomposes as

$$\mathbb{R}^6 = \operatorname{range}(\boldsymbol{C}) \oplus \ker(\boldsymbol{C}^{\mathsf{T}}). \tag{8}$$

The orthogonal projectors onto these subspaces are

$$\boldsymbol{P}_R = \frac{1}{3}\boldsymbol{C}\boldsymbol{C}^\mathsf{T}, \quad \boldsymbol{P}_N = \boldsymbol{I}_6 - \frac{1}{3}\boldsymbol{C}\boldsymbol{C}^\mathsf{T}. \tag{9}$$

Using $\boldsymbol{C}^\mathsf{T}\boldsymbol{C} = 3\boldsymbol{I}_2$,

$$\boldsymbol{P}_R^2 = \frac{1}{9}\boldsymbol{C}(\boldsymbol{C}^\mathsf{T}\boldsymbol{C})\boldsymbol{C}^\mathsf{T} = \boldsymbol{P}_R, \quad \boldsymbol{P}_R^\mathsf{T} = \boldsymbol{P}_R. \tag{10}$$

It follows that $\boldsymbol{P}_N^2 = \boldsymbol{P}_N$, $\boldsymbol{P}_N^\mathsf{T} = \boldsymbol{P}_N$, $\boldsymbol{P}_R\boldsymbol{P}_N = \boldsymbol{0}$, and $\boldsymbol{C}^\mathsf{T}\boldsymbol{P}_N = \boldsymbol{0}$. One explicit basis of $\ker(\boldsymbol{C}^\mathsf{T})$ is

$$\boldsymbol{N} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}, \quad \boldsymbol{C}^\mathsf{T}\boldsymbol{N} = \boldsymbol{0}. \tag{11}$$

The first three columns represent equal common-mode changes within each antagonistic tendon pair. The fourth column extends the three-pair vector $\boldsymbol{n}_a = [1, -1, 1]^\mathsf{T}$ to the full six-tendon coordinate space. Each basis direction is rejected by the algebraic curvature reconstruction. The physical effect of a tendon motion also depends on transmission, loading and pretension.

For an arbitrary measured vector $\boldsymbol{q}_m$,

$$\boldsymbol{q}_m = \boldsymbol{P}_R\boldsymbol{q}_m + \boldsymbol{P}_N\boldsymbol{q}_m, \quad \hat{\boldsymbol{c}} = \frac{1}{3dL}\boldsymbol{C}^\mathsf{T}\boldsymbol{q}_m, \quad \boldsymbol{r}_q = \boldsymbol{P}_N\boldsymbol{q}_m. \tag{12}$$

The representable part is $\boldsymbol{P}_R\boldsymbol{q}_m = dL\boldsymbol{C}\hat{\boldsymbol{c}}$, while $\boldsymbol{r}_q$ is rejected by the ideal curvature estimator. An in-range bias $dL\boldsymbol{C}\boldsymbol{b}_c$ has zero residual and produces the curvature bias $\boldsymbol{b}_c$, which is why a small residual is not a complete model-validation test.

All tendon velocities that give the ideal reference curvature velocity $\dot{\boldsymbol{c}}_r$ are

$$\dot{\boldsymbol{q}} = dL\boldsymbol{C}\dot{\boldsymbol{c}}_r + \boldsymbol{P}_N\boldsymbol{z}, \quad \boldsymbol{z} \in \mathbb{R}^6. \tag{13}$$

The two terms are orthogonal. Therefore,

$$\|\dot{\boldsymbol{q}}\|_2^2 = 3d^2L^2\,\|\dot{\boldsymbol{c}}_r\|_2^2 + \|\boldsymbol{P}_N\boldsymbol{z}\|_2^2\,,$$

The minimum is attained precisely when $\boldsymbol{P}_N\boldsymbol{z} = \boldsymbol{0}$, giving the unique minimum-Euclidean-norm velocity $\dot{\boldsymbol{q}} = dL\boldsymbol{C}\dot{\boldsymbol{c}}_r$. Setting $\boldsymbol{z} = \boldsymbol{0}$ is one parameter choice; the parameter is not unique because any $\boldsymbol{z} \in \mathrm{range}(\boldsymbol{C})$ has the same zero projection. The statement concerns the algebraic velocity norm and does not imply minimum energy, positive tension, slack prevention or physical feasibility of other null-space motions.

2 Explicit tendon and reel commands

Define

$$v_x = \dot{c}_{d,x} - \frac{\dot{B}_x}{B_x}e_x + k_x e_x, \tag{14}$$

$$v_y = \dot{c}_{d,y} - \frac{\dot{B}_y}{B_y}e_y + k_y e_y. \tag{15}$$

Then $\dot{\boldsymbol{c}}_r = [v_x, v_y]^\mathsf{T}$, and the six entries of $\dot{\boldsymbol{q}} = dL\boldsymbol{C}\dot{\boldsymbol{c}}_r$ are

$$\begin{aligned}
\dot{q}_1 &= dLv_x, \\
\dot{q}_2 &= dL\left(\tfrac{1}{2}v_x + \tfrac{\sqrt{3}}{2}v_y\right), \\
\dot{q}_3 &= dL\left(-\tfrac{1}{2}v_x + \tfrac{\sqrt{3}}{2}v_y\right), \\
\dot{q}_4 &= -dLv_x, \\
\dot{q}_5 &= dL\left(-\tfrac{1}{2}v_x - \tfrac{\sqrt{3}}{2}v_y\right), \\
\dot{q}_6 &= dL\left(\tfrac{1}{2}v_x - \tfrac{\sqrt{3}}{2}v_y\right).
\end{aligned} \tag{16}$$

They obey

$$\dot{q}_4 = -\dot{q}_1, \quad \dot{q}_5 = -\dot{q}_2, \quad \dot{q}_6 = -\dot{q}_3, \quad \dot{q}_1 - \dot{q}_2 + \dot{q}_3 = 0. \tag{17}$$

The total and component rate bounds follow from $\boldsymbol{C}^{\mathsf{T}}\boldsymbol{C} = 3\boldsymbol{I}_2$:

$$\|\dot{\boldsymbol{q}}\|_2 = \sqrt{3}\, dL \, \|\dot{\boldsymbol{c}}_r\|_2 , \quad |\dot{q}_i| \le dL \, \|\dot{\boldsymbol{c}}_r\|_2 . \tag{18}$$

Because $q_i = l_{i,\mathrm{ref}} - l_i$ and the references are constant,

$$\dot{l}_i = -\dot{q}_i. \tag{19}$$

If reel i has calibrated effective radius $r_{s,i} > 0$ and measured winding sign $\sigma_i \in \{-1, 1\}$, a constant-radius interface is

$$\omega_i = \sigma_i \frac{\dot{q}_i}{r_{s,i}}. \tag{20}$$

Both $r_{s,i}$ and σ_i are calibration quantities. Layered winding can make the effective radius state dependent, in which case Eq. (20) must be replaced by a measured nonlinear conversion. None of these relations enforces positive tendon tension.

3 Straight-configuration series

Although the main-text constant-curvature map contains terms involving $1/\kappa$, the apparent singularity at $\kappa = 0$ is removable. To evaluate the straight configuration, consider the Taylor expansions

$$\begin{aligned}
A(\kappa) &= \frac{L^2}{2} - \frac{L^4}{24}\kappa^2 + \frac{L^6}{720}\kappa^4 + O(\kappa^6), \\
G(\kappa) &= L - \frac{L^3}{6}\kappa^2 + \frac{L^5}{120}\kappa^4 + O(\kappa^6), \\
A'(\kappa) &= -\frac{L^4}{12}\kappa + \frac{L^6}{180}\kappa^3 + O(\kappa^5), \\
G'(\kappa) &= -\frac{L^3}{3}\kappa + \frac{L^5}{30}\kappa^3 + O(\kappa^5).
\end{aligned} \tag{21}$$

For numerical implementation, the Taylor series expansions are used when $L\kappa$ is sufficiently small to avoid numerical cancellation. This numerical treatment does not alter the analytical tip-position map. Using $\kappa^2 = c_x^2 + c_y^2$, the corresponding multivariate expansions of the tip-position components are

$$\begin{aligned}
p_x &= \frac{L^2}{2}c_x - \frac{L^4}{24}c_x(c_x^2 + c_y^2) + O(\|\boldsymbol{c}\|^5), \\
p_y &= \frac{L^2}{2}c_y - \frac{L^4}{24}c_y(c_x^2 + c_y^2) + O(\|\boldsymbol{c}\|^5), \\
p_z &= L - \frac{L^3}{6}(c_x^2 + c_y^2) + \frac{L^5}{120}(c_x^2 + c_y^2)^2 + O(\|\boldsymbol{c}\|^6).
\end{aligned} \tag{22}$$

Differentiating these expansions gives

$$\begin{aligned}
\frac{\partial p_x}{\partial c_x} &= \frac{L^2}{2} - \frac{L^4}{24}(3c_x^2 + c_y^2) + O(\|\boldsymbol{c}\|^4), \\
\frac{\partial p_x}{\partial c_y} &= -\frac{L^4}{12}c_x c_y + O(\|\boldsymbol{c}\|^4), \\
\frac{\partial p_y}{\partial c_x} &= -\frac{L^4}{12}c_x c_y + O(\|\boldsymbol{c}\|^4), \\
\frac{\partial p_y}{\partial c_y} &= \frac{L^2}{2} - \frac{L^4}{24}(c_x^2 + 3c_y^2) + O(\|\boldsymbol{c}\|^4), \\
\frac{\partial p_z}{\partial c_x} &= -\frac{L^3}{3}c_x + O(\|\boldsymbol{c}\|^3), \\
\frac{\partial p_z}{\partial c_y} &= -\frac{L^3}{3}c_y + O(\|\boldsymbol{c}\|^3).
\end{aligned} \tag{23}$$

4 Related control formulations

Table S1: Related control approaches and the scope of the present comparison. The rows summarize reported formulations, not a common-platform performance ranking.

Reference	Object and approach	Boundary and time meaning	Input or uncertainty treatment	Reported evidence
Zhu and Li [1]	Uncertain nonlinear system; composite fuzzy control	Quartic performance function reaches its terminal width at assigned T_e	Dead-zone inputs and uncertain nonlinearities; learning/adaptation	Numerical example
Xu [2]	Continuum-robot dynamics; adaptive fault-tolerant control	Asymmetric time-varying BLF bounds transient and steady tracking; no fixed-cutoff claim is used in this comparison	Function approximation, disturbance observer and actuator-fault model	Simulation and experiment
Zhang et al. [3]	Lower-limb exoskeleton; transformed performance control	Specified tracking region within a preset time	Time-varying state constraints; saturation treated using a hyperbolic-tangent formulation	Numerical example; different mechanism
Ba et al. [4]	Tendon-driven continuum robots; analytical differential feedback	Positioning and path tracking; not used as a prescribed-time boundary benchmark	Tension supervision addresses slack	Experiments on three robot structures
Pan et al.; He et al. [5, 6]	Continuum visual tracking; estimated Jacobian and QP control	Visual-feature tracking objective rather than the error envelope evaluated here	Physical/control constraints; robust estimation or RNN-based QP solver	Simulation and experiment
Present study	Six-tendon nominal curvature kinematics; explicit feedback	Cubic boundary reaches nonzero ε by selected T; subsequent nominal error decay	Rate/travel/tension effects evaluated separately; nominal theorem assumes accurate execution	Ideal checks, MuJoCo, four-channel scalar visual-feedback experiment

Only proportional baselines with shared settings are evaluated as direct controller comparisons. The quartic boundary experiment changes the boundary profile within the same feedback structure; it does not reproduce Zhu and Li's fuzzy-learning controller. The explicit control law requires no online optimization.

5 Complete archived one-at-a-time table

Table S2: One-at-a-time perturbation results. Error/RMS columns use $t \geq T_c$; minimum tension, observed slack duration and saturation use the full record. Saturation is the mean of six actuator flags, expressed as a percentage.

Condition	$\max_j \lvert\xi_j\rvert$ $(t \geq T_c)$	Tip RMS, $t \geq T_c$ (mm)	δ_{CC} RMS, $t \geq T_c$ (mm)	Min. tension (N)	Slack (s)	Saturation (%)	Violations
Damping -30%	0.0008	0.3682	0.3696	1.7903	0.0	0.0	0
Damping +30%	0.0008	0.3667	0.3688	2.8136	0.0	0.0	0
Radius model -10%	0.0017	0.3659	0.369	2.8136	0.0	0.0	0
Radius model +10%	0.0006	0.3679	0.3692	2.8136	0.0	0.2661	0
Two-cycle delay	0.0006	0.3677	0.3692	2.8136	0.0	0.0	0
Gravity	0.1358	28.0331	28.4599	3.4356	0.0	0.0	0
Tendon gain -20%	0.0009	0.3663	0.3687	3.2536	0.0	0.0	0
Tendon gain +20%	0.0008	0.368	0.3694	2.3736	0.0	0.0	0
Measurement noise	0.0312	0.369	0.3868	2.8132	0.0	0.0	0
Pretension -20%	0.0008	0.3685	0.3696	1.7728	0.0	0.0	0
Pretension +20%	0.0025	0.3942	0.3825	3.8503	0.0	0.0	0
Young's modulus -20%	0.002	0.3875	0.3786	2.8486	0.0	0.0	0
Young's modulus +20%	0.0008	0.3684	0.3696	2.7755	0.0	0.0	0

6 Tip-position derivatives and analytical parameter roles

6.1 Jacobian and rotational symmetry

For the main-text tip map, main-text Eq. (36), and Jacobian definition, main-text Eq. (37), the chain-rule calculation is as follows. Since $\boldsymbol{p}$ depends on $\boldsymbol{c}$ both explicitly and implicitly through $\kappa = \|\boldsymbol{c}\|$, the Jacobian is obtained by the chain rule. For $\kappa > 0$,

$$\frac{\partial \kappa}{\partial \boldsymbol{c}} = \frac{\boldsymbol{c}^{\mathsf{T}}}{\kappa}, \quad \frac{\partial \kappa}{\partial c_x} = \frac{c_x}{\kappa}, \quad \frac{\partial \kappa}{\partial c_y} = \frac{c_y}{\kappa}. \tag{24}$$

The scalar derivatives of $A(\kappa)$ and $G(\kappa)$ are

$$A'(\kappa) = \frac{L\kappa \sin(L\kappa) - 2[1 - \cos(L\kappa)]}{\kappa^3}, \tag{25}$$

$$G'(\kappa) = \frac{L\kappa \cos(L\kappa) - \sin(L\kappa)}{\kappa^2}. \tag{26}$$

Consequently, for $\kappa > 0$, the 3×2 tip-position Jacobian is

$$\boldsymbol{J}_p(\boldsymbol{c}) = \begin{bmatrix} A(\kappa) + \dfrac{A'(\kappa)}{\kappa} c_x^2 & \dfrac{A'(\kappa)}{\kappa} c_x c_y \\ \dfrac{A'(\kappa)}{\kappa} c_x c_y & A(\kappa) + \dfrac{A'(\kappa)}{\kappa} c_y^2 \\ \dfrac{G'(\kappa)}{\kappa} c_x & \dfrac{G'(\kappa)}{\kappa} c_y \end{bmatrix}. \tag{27}$$

For $\boldsymbol{R} \in \mathrm{SO}(2)$ and $\widetilde{\boldsymbol{R}} = \mathrm{blkdiag}(\boldsymbol{R}, 1)$, radial symmetry gives

$$\boldsymbol{p}(\boldsymbol{R}\boldsymbol{c}) = \widetilde{\boldsymbol{R}}\boldsymbol{p}(\boldsymbol{c}), \qquad \boldsymbol{J}_p(\boldsymbol{R}\boldsymbol{c}) = \widetilde{\boldsymbol{R}}\boldsymbol{J}_p(\boldsymbol{c})\boldsymbol{R}^{\mathsf{T}}. \tag{28}$$

Both rotation matrices are orthogonal, hence

$$\|\boldsymbol{J}_p(\boldsymbol{R}\boldsymbol{c})\|_2 = \|\boldsymbol{J}_p(\boldsymbol{c})\|_2 . \tag{29}$$

The numerical assessment on a circular curvature domain can therefore be reduced to a search over curvature magnitude. The straight-state series are given in Supplementary Note 3.

6.2 Performance specifications

The nominal performance specifications are

$$\begin{aligned} |e_j(t)| &< B_j(t) \quad (t \geq 0), \\ |e_j(t)| &< \varepsilon_j \quad (t \geq T_j), \\ \lim_{t\to\infty} e_j(t) &= 0. \end{aligned} \tag{30}$$

The second requirement is entry into a nonzero terminal band by T_j; the third is subsequent asymptotic convergence under nominal accurate execution.

6.3 Parameter and residual roles

Table S3: Parameter and residual roles in the analysis and implementations.

Quantity	Units	Source or role	Applicable check
$L, d, \boldsymbol{C}$	m, m, dimensionless	Nominal geometry and tendon numbering	Model/configuration identity; signs; $\boldsymbol{C}^{\mathsf{T}}\boldsymbol{C} = 3\boldsymbol{I}_2$
$B_{j0}, \varepsilon_j, T_j, k_j$	$\mathrm{m}^{-1}, \mathrm{m}^{-1}, \mathrm{s}, \mathrm{s}^{-1}$	Performance design	Initial admissibility; task tolerance; rate/travel/sampling checks
g, b	Curvature per command degree; curvature	Local four-channel hardware calibration	Recorded command levels; repeats; target extrapolation; not six-tendon geometry
n_j, η_j	m^{-1}	Sensing error and a bound, if available	Uncertainty at the actual signal level; statistical spread is not a deterministic bound
$d_j, \bar{\delta}_j$	$\mathrm{m}^{-1}\mathrm{s}^{-1}$	Actual and effective execution-rate deviation	Platform-specific execution evidence; no inferred all-time bound from sample slopes
Shape-fit residual	m	Fitted backbone versus measured/model nodes	Actual fit objective, node selection and sample window
δ_{CC}	m	Fitted-curve tip versus MuJoCo tip	Retained separately in Eq. (76)

7 Numerical model and evaluation methods

7.1 Fixed-time discretization comparison

All controller comparisons use the specified $N = 16$ model. The separate link-discretization comparison uses the original model parameterization for each N and a common fixed-time protocol. Its within-model shape residual is defined by main-text Eq. (77); the fitted set contains the base and the N link centers, while the terminal tip is excluded. The tip and curvature difference columns compare model endpoints and fitted curvature vectors with the $N = 24$ snapshot. A comparator therefore has zero inter-model differences but can have a nonzero constant-curvature fitting residual.

Table S4: Fixed-time, fixed-protocol comparison of the supplied link discretizations: 2-s pretension followed by a 3-s command hold at the 2-ms physics step. Tip and curvature differences use the $N = 24$ snapshot; shape RMSE is the within-model fitting residual. Timings are the recorded original real-time factors.

N	Tip diff. (mm)	$\Delta\kappa$ (m^{-1})	Shape RMSE (mm)	Real-time factor
8	1.7066	0.0237	0.3468	82.0508
12	1.1237	0.0379	0.3654	54.2026
16	1.0588	0.0214	0.125	36.2731
24	0.0	0.0	0.6263	19.0938

The $N = 24$ residual at this sample was 0.626330 mm, with a generalized-velocity norm of 11.18 rad/s. At the same time with a 1-ms step, its residual was 0.043686 mm. Increasing the fit iteration cap did not change the original four-evaluation solution. A common longer-hold comparison used 1-ms and 0.5-ms physics steps with 19-s and 20-s endpoint checks. The $N = 12$ model retained step dependence: its 20-s tips differed by 6.2405 mm between steps, and its tips changed by 0.56–0.58 mm from 19 to 20 s. Supplementary Data Tables D15–D17 report the individual results. These finite-time comparisons characterize fitting and implementation sensitivity; they do not supply a steady mesh-convergence limit for all the supplied models.

7.2 State timing and step sensitivity

The control implementation reads the derived body poses after `mj_step`, whose integration advances the state after its forward calculations. Its feedback pose cache has one physics-step age. Synchronized recorded nodes are evaluated in an independent data object copied from the current state; those nodes supply the offline shape fits. The controller remains unchanged in the comparisons.

Eleven task/controller pairs were repeated at a 1-ms physics step with the 20-ms control period retained. Their entry-time changes were at most 3 ms, the maximum absolute J_q change was 0.023%, and the largest shared-time tip difference was 1.63 mm. Supplementary Data Table D26 gives the maximum and RMS tip differences, deadline differences, normalized maxima and costs for each pair. Because step refinement also changes the pose-cache age from 2 to 1 ms, these pairs assess the combined numerical and feedback-timing implementation.

7.3 Task, cost and perturbation definitions

E7 has 28 requirements per method, implemented in 84 independent 10-s simulations. The task gate uses the recorded control-sample maximum, a separately fitted synchronized state at the exact deadline, and all saved 2-ms tendon tensions. The original numerical margin is $\tau = 10^{-3}$; strict zero-margin re-evaluation gives the same task counts. All 5001 synchronized physical states per run were fitted for dense evaluation. Identical complete geometry sequences share the same deterministic fitting result, while each run is evaluated using its own time, boundary and tension criterion. Complete sampled/strict/dense comparisons are listed in Data Table D22.

The new-task J_q is the sum of squared applied rates multiplied by their executed 20-ms intervals. The terminal mathematical evaluation at 10 s has zero duration. The rate-clipping duration counts the union of command intervals affected in at least one tendon. For the original E4 comparison, the published effort metric instead uses the trapezoidal rule on its saved applied-rate sequence. Both definitions have units m^2/s. The unrounded E4 integrals give a 4.41% increase for the proposed method; Data Table D20 supplies the complete values.

In the original E4 logs, the applied-rate sample at t_k records the preceding control interval, whereas E7 records the rate held over $[t_k, t_{k+1})$. For the $T = 5$ s comparison, the 500 executed rate vectors agree exactly after this one-sample alignment; the trapezoidal E4 and held-command E7 costs therefore agree to the

displayed precision because the original endpoint rates are zero or negligible. For a fixed uniformly sampled sequence, $J_L - J_T = \Delta t(f_0 - f_N)/2$, where $f_k = \|\dot{\boldsymbol{q}}_k\|_2^2$.

The original E5 results report curvature-envelope tracking and separate actuation diagnostics over its protocol. The paired sensitivity criterion additionally requires the exact-deadline and 0.05-N tension gates. Envelope, deadline, later-band, tension and estimation reasons can overlap; Data Tables D23–D25 distinguish these reasons and definitions. The 56 original benchmark records, E6 application, 84 E7 tasks, two additional quartic runs, 60 paired perturbations and repeated/half-step checks are counted separately.

7.4 Straight-state reference functions

For the ideal Python case in Figure 4(c), the stored reference is $c_{d,x} = 4\sin(2\pi t/8)$ m^{-1} and $c_{d,y} = 0$ over $0 \le t \le 12$ s, with initial curvature $(0.6, 0.2)^{\mathsf{T}}$ m^{-1}. Reference zeros occur at 0, 4, 8 and 12 s; only 4 and 8 s are internal reference crossings. E3 in the physical model uses $c_{d,x} = 3\sin(2\pi t/4)$ m^{-1} and $c_{d,y} = 0$ over the same interval, giving reference zeros at 0, 2, 4, 6, 8, 10 and 12 s. The five zeros in the open interval are internal crossings. These are separate tasks, and the reference zeros are distinguished from the measured/fitted trajectory's zero times.

8 Visual-feedback hardware methods

Markers and metric coordinates. R0 and R1 are the left and right backboard markers. Their initial stable positions define a two-point similarity reference; each current image is mapped to this reference by translation, rotation and isotropic scale correction. The upper segment is observed at B0 (base), U1, U2 and J (junction), and the lower segment at J, L1, L2 and T (tip). The lower segment is monitored but is not used as the controlled curvature. The configured B0–T distance is 290 mm. Its initial stabilized pixel distance fixes the metric scale for the session; the logged scale is constant within each retained trial. The configured R0–R1 distance of 289 mm is not used by this implementation to set the pixel scale. These values are recorded setup measurements, rather than a recovered intrinsic camera calibration.

Coordinate smoothing and signed arc fitting. Detected marker centers are smoothed as $p_f[n] = 0.42p_d[n] + 0.58p_f[n-1]$ before reference stabilization and arc fitting. For a segment, a signed planar constant-curvature centerline is anchored at its first marker. Its tangent direction θ, length L and bend angle $\beta = \kappa L$ are estimated by bounded nonlinear least squares, using SciPy's trust-region reflective solver and the `soft_l1` loss with a 1-mm residual scale. The two internal marker locations are initially estimated jointly as ordered arc fractions; equally spaced stickers are not assumed. The endpoints have fractions 0 and 1. After 20 valid estimates, the componentwise median fractions are fixed if adjacent fractions differ by more than 0.01 and their internal median absolute deviation is at most 0.025. Five consecutive invalid fits trigger a new fraction initialization. Length is estimated rather than assigned a fixed upper-segment value. The stored CSV contains signed curvature, marker coordinates and fit RMSE, but does not retain each fitted length, tangent direction or frozen internal fraction.

The centerline uses the analytic straight limit of $\sin(\kappa s)/\kappa$ and $(1-\cos(\kappa s))/\kappa$. A fit must converge with finite parameters and RMSE below $\min(3\text{ mm}, 0.03L)$. The runtime-valid gate additionally requires nine detected markers, the prescribed marker order, valid upper and lower fits, and camera-reference changes within 5 pixels of translation, 0.5° of rotation and 1% of scale. All 1990 stored tracking frames in the retained trials satisfy this complete gate; all nine marker statuses are `detected`.

Three signal levels and their clocks. The logged marker coordinates are EMA coordinates, not the unsmoothed detector centers. The historical fields `kappa_upper_raw` and `kappa_upper_filtered` are aliases: the recorder writes the same per-frame fit into both fields, and their saved values agree exactly. The controller reads a separate median of valid fits in the 0.2-s interval ending at the latest processed camera timestamp. It subtracts the frozen reference $\kappa_0 = 1.305095121\ \text{m}^{-1}$ to obtain the signed scalar feedback c_h.

Vision timestamps are host-monotonic frame-arrival times relative to the recorder origin; they are not camera exposure timestamps. `COMMAND_ONSET` uses the recorder clock, whereas control timestamps are relative to the immediately following controller start. The start offset, controller-read time, processing-finish time and physical actuation time are not separately logged. Frame-level comparisons therefore use the recorded command-onset convention, without shifting the visual curve. The latest-frame processing loop may skip camera arrivals; unique recorded frame identifiers do not establish that every captured camera frame was processed. No duplicate recorded timestamps, frame identifiers or video-frame indices were found in the 1990 tracking records, and no adjacent stored marker-coordinate sets were identical.

Boundary and sample metrics. Before each trial, a 3-s baseline sequence of recent-window curvature medians defines the initial relative curvature. The runtime takes σ as the larger of its sample SD and the frozen static SD of $0.03556169\ \text{m}^{-1}$, then sets $B_0 = 1.25|e_0| + 4\sigma$ and $\varepsilon = \max(4\sigma, 0.10|c_{h,d}|)$. The resulting per-trial B_0 and σ are recorded; the baseline query sequence is not separately stored and is distinguished from the first control sample.

For every trial, the recorded frozen B_0, ε and T are used to evaluate the cubic boundary at each frame's actual time relative to `COMMAND_ONSET`. The error is $e_h = c_{h,d} - (\kappa_{\text{upper,fit}} - \kappa_0)$, and the normalized error is $e_h/B(t)$. Only $0 \le t < 12$ s before `STEADY_END` is included. Return motion starts after 14 s in all six trials and contributes no tracking frame. The complete validity gate above is retained. A boundary exceedance is a recorded sample with $|e_h|/B \ge 1$.

Each trial contains 24 control samples. Terminal RMSE is the unweighted root mean square of those original control errors whose timestamps satisfy $t \ge 8$ s. Terminal entry is the first recorded sample after which all subsequent samples of that same signal level remain within $|e_h| \le \varepsilon$ for the remainder of its recorded tracking support. Means and sample SDs are computed across the three independent trials per controller, with SD denominator $n-1$. The 32.5% reduction is computed from the unrounded per-controller mean terminal RMSEs; per-frame fits are not substituted for this outcome.

For display, main-text Figure 16(b,c) linearly interpolates each of the three feedback sequences on a 0.1-s grid restricted to their common recorded support. The interpolated values contribute to the plotted mean

and SD only. The normalized-error curves end at their last actual control sample; no feedback value is appended at 12 s. Limit-state time fractions, when reported, describe the held commanded target over its update intervals through 12 s, and are distinct from a new feedback observation.

Calibration, attempts and return checks. The six calibration responses are the median signed upper-segment curvature in the final second before `RETURN_START`, minus the mean in the recorded baseline interval. The two input levels are 10 and 15 command degrees, with three independent trials at each level. Fitting $c_h = gu + b$ gives $g = -0.05574746$ m^{-1} per command degree, $b = -0.02460074$ m^{-1} and $R^2 = 0.9978953$. The target $c_{h,d} = -0.46506248$ m^{-1} corresponds to $(c_{h,d} - b)/g = 7.90102$ command degrees, below the directly tested interval. The feedback implementation uses g for incremental command conversion; it does not command this static-equivalent value as an open-loop target.

The formal series contains eight attempts: six retained, one aborted before tracking began because of visual-marker loss, and one completed but excluded for processing below 15 frames/s. Four pilot attempts are listed separately, including two protocol-related aborts. The acceptance-threshold snapshot records pre-existing software defaults but is not evidence of prospective registration. A runtime return check uses the latest 0.5-s median at the return observation event; the offline return residual uses the final 1-s mean of the complete recording. These different windows explain why a runtime return warning and the later residual are reported separately. Neither is used alone to reclassify tracking performance.

8.1 Local calibration-slope sensitivity

For a constant target, the effect of a constant local slope error can be isolated by writing $g_{\text{true}} = \gamma \hat{g}$, $\gamma > 0$, and $\dot{u} = \dot{c}_r/\hat{g}$. With exact sensing and an unclipped continuous command,

$$\dot{e} = -\gamma \left(k - \frac{\dot{B}}{B} \right) e, \qquad \dot{\xi} = \left[-\gamma k + (\gamma - 1) \frac{\dot{B}}{B} \right] \xi. \tag{31}$$

The corresponding solutions are

$$e(t) = e(0) e^{-\gamma k t} \left(\frac{B(t)}{B_0} \right)^{\gamma}, \qquad \xi(t) = \xi(0) e^{-\gamma k t} \left(\frac{B(t)}{B_0} \right)^{\gamma - 1}. \tag{32}$$

Thus slope mismatch changes the normalized transient, while the terminal constant-boundary phase has decay rate γk. An underestimated response magnitude ($\gamma > 1$) accelerates the ideal correction; an overestimated magnitude ($0 < \gamma < 1$) weakens it. The local gain ratio remains to be identified near the target; this sensitivity model specifies analytical cases rather than a measured calibration interval.

For $\gamma \geq 1$, $\dot{B}/B \leq 0$ makes the normalized coefficient at most $-\gamma k$. For $0 < \gamma < 1$, relative boundary contraction contributes positively to that coefficient; the closed-form expression gives the appropriate transient check. A simple sufficient uniform condition is $|\xi(0)| \, (B_0/\varepsilon)^{1-\gamma} < 1$ in the latter case. Once $B = \varepsilon$, both errors decay exponentially at rate γk.

For the matched proportional law under the same local assumptions, $e_{\text{B1}}(t) = e(0) e^{-\gamma k t}$ and $\xi_{\text{B1}}(t) = \xi(0) e^{-\gamma k t} B_0 / B(t)$. This comparison separates the calibration-slope factor from the prescribed boundary compensation. The actuator-force gains varied in the tendon-driven MuJoCo model are different physical quantities from this scalar local slope.

8.2 Command protocol and interpretation

The scalar command grid is 0.1 motor-command degree, with 0–12-degree travel and a 0.5-degree maximum change from the previous target. The protocol selector chooses a reachable payload that avoids a premature frame terminator, resolving an exact tie toward lower magnitude. The requested, controller-limited and encoded targets and the rate/position/protocol fields were recorded; separate step-clipping and quantization incidences were reconstructed offline. Time-weighted limit fractions describe held target commands through the tracking interval; they are distinct from newly measured feedback. Encoder records consist of a pre-motion snapshot and a post-return snapshot, while the IMU and lower-section vision are diagnostic signals. Tendon tensions were not measured in these trials.

9 AI-assisted figure preparation and traceability

Tools and scope. OpenAI Codex, provided by OpenAI (model `gpt-5.6-sol`, as confirmed by Yi Lu), assisted with figure-source code, plotting, extraction and layout. The current figure sources use TikZ or deterministic Python/Matplotlib code and recorded photographs, video crops or numerical arrays. No current figure asset was identified as a direct text-to-image output in the inspected sources and available records. This finding is bounded by the retained records.

Figures 1–3. Codex assisted with the TikZ source and layout of the performance-envelope schematic (Figure 1), geometry/coordinate schematic (Figure 2) and controller-flow schematic (Figure 3). Their editable sources are `fig_02_vector_envelope.tex`, `fig_01_combined.tex` with `fig_01a_geometry_3d.tex`, and `fig_03_control_compact.tex`, respectively. These are code-rendered diagrams, not generated experimental images.

Figure 4, Figures 5–12 and Data Figures D1–D4. Figure 4 uses AI-assisted Python plotting code in `redraw_ideal_summary.py`. Figures 5–12 and Data Figures D1–D4 use `render_simulation_figures.py`. The plotted inputs are the archived analytical/numerical outputs, simulation trajectories and evaluation tables identified in the figure-source ledger. AI assistance concerned code and presentation; it did not replace the executed simulation records with synthetic research data. The source ledger maps each figure separately to its SVG/PDF or PNG output and underlying inputs.

Figure 13. The platform diagram combines photographs taken by Yi Lu with programmatically drawn host, phone and IMU outlines, arrows, labels and the recorded visual-diagnostic inset. Codex assisted with the Python layout and drawing code in `redraw_architecture_round4.py`. The photographic portions remain recorded experimental photographs; the generic outlines are Matplotlib primitives rather than imported stock icons or text-to-image assets. The output is `Figure11_Hardware_Platform_and_Measurement.pdf`, with an editable SVG and generator.

Figure 14. The five panels are crops from the experimental video recorded by Yi Lu. AI-assisted code supports frame extraction, cropping and LaTeX montage layout; it does not synthesize the arm or camera scene. The retained original frame indices are 746, 883, 895, 912 and 929, with crop rectangle $(230, 70, 730, 720)$ in pixel coordinates. The editable montage is `F14_layout.tex`; `extract_F14_frames.py` and the five PNG outputs are retained. These operations reproduce the selected recorded image pixels.

Figures 15–16 and Data Figure D5. Codex assisted with Python plotting and typography through `typography_hardware_round3.py` and the retained `frozen_r2_render_hardware_figures.py`. The source arrays and tables are the recorded hardware/calibration and offline audit records. The plotted curves, sample support and existing scientific interpretation are retained.

Graphical Abstract. Codex assisted with deterministic drawing and layout code in `graphical_abstract_generator.py`. The six-tendon geometry, contracting boundary and controller blocks are programmatic schematic elements. The retained outputs are the PDF, SVG and PNG graphical abstract, together with the generator. This is a code-rendered schematic, not a text-to-image scientific image.

Prompt and output records. The accompanying `AI_FIGURE_DISCLOSURE.md` and `FIGURE_AI_LEDGER.csv` identify all 22 figure items, model evidence, editable sources, outputs and rights questions. The `FIGURE_TOOL_TRACE_INDEX.json` records available source-edit/run references using session identifiers, timestamps, line numbers and argument hashes; these are traceable assistance records, not reconstructed prompts. A complete set of the original per-figure input prompts and intermediate model responses has not been retained and is not recreated retrospectively. The source archive retains the edited figure code and outputs with SHA-256 manifests. Original full video and private conversation logs are excluded from that archive. Availability of these records does not constitute a journal ruling on AI-use eligibility or copyright.

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Supplementary numerical and experimental data

Prescribed-Time Contracting-Boundary Control of a Tendon-Driven Flexible Arm

This appendix reports the nominal task grid, boundary-profile comparisons, perturbation outcomes, hardware observations and numerical sensitivity checks. Tables D1-D26 and Figures D1-D5 retain the recorded outcomes, including unsuccessful cases and unresolved cross-step differences.

Execution and observation units

Evidence group	Count	Interpretation
Original numerical campaign	56	Historical E0-E5 records; kept separate
P0 reproduction	4	Unchanged archived-code reproductions
Comparison executions	168	84 grid + 2 ablations + 60 paired perturbations + 11 repeats + 11 half-step checks; separate discretization diagnostics are not pooled
Representative dense validation	33 runs	220,033 representative-record fits; prior coverage
Complete nominal dense check	84 records	420,084 run-level states; 48 exact geometry sequences, including 6 verified prior-fit and 42 newly fitted sequences
Hardware formal comparison	6 trials	Three per controller; 144 recorded control updates
Hardware frame-level audit	1,990 samples	80 envelope exceedances in five trials; a different signal layer

P, B1 and B2 in the numerical sections denote the cubic controller, the matched proportional baseline and the task-tuned constant-gain baseline. In the historical hardware section, P1-P3 and B1-B3 are trial identifiers; hardware B2 is the second baseline trial, not the task-tuned numerical baseline.

The data tables use independent numbering D1-D26. They supplement, and do not renumber, the mathematical notes and Tables S1-S4. Units and observation levels are stated at each table. Values are rounded only for display; the underlying research records retain full precision.

The underlying data are available from the corresponding author upon reasonable request, subject to applicable permissions and restrictions, as stated in the manuscript. The supplementary source package includes editable display tables and DATA_INDEX.csv; full-precision records are retained separately.

Data index and precision

Nominal comparisons and profile ablations: D1-D7 and D22. Hardware trials, calibration, signal levels and attempts: D8-D14. Discretization and metric checks: D15-D21 and D26. Perturbation definitions and failure categories: D23-D25. DATA_INDEX.csv maps record groups to the included display-table files. These display files preserve the precision shown here; calculations use the retained unrounded records.

Complete nominal grid: P

Data Table D1. All 28 task requirements for P

T (s)	epsilon (m^-1)	Success (1/0)	Xmax control	Exact-T error/epsilon	10^3 Jq (m^2/s)
0.75	0.05	1	0.692820	0.106513	1.459248
0.75	0.1	1	0.692820	0.168003	1.449466
0.75	0.2	1	0.692820	0.211215	1.429836
0.75	0.4	1	0.692820	0.231567	1.394474
1	0.05	1	0.692820	0.068537	1.290988
1	0.1	1	0.692820	0.089897	1.285097
1	0.2	1	0.692820	0.107467	1.273194
1	0.4	1	0.692820	0.121186	1.249415
1.25	0.05	1	0.692820	0.054054	1.145266
1.25	0.1	1	0.692820	0.065043	1.140375
1.25	0.2	1	0.692820	0.074375	1.130620
1.25	0.4	1	0.692820	0.082228	1.111259
1.5	0.05	1	0.692820	0.043501	1.040531
1.5	0.1	1	0.692820	0.049042	1.037042
1.5	0.2	1	0.692820	0.053907	1.030085
1.5	0.4	1	0.692820	0.058162	1.016268
2	0.05	1	0.692820	0.026043	0.924711
2	0.1	1	0.692820	0.026723	0.922685
2	0.2	1	0.692820	0.027574	0.918641
2	0.4	1	0.692820	0.028531	0.910599
3	0.05	1	0.692820	0.009696	0.831095
3	0.1	1	0.692820	0.008506	0.830162
3	0.2	1	0.692820	0.007736	0.828298
3	0.4	1	0.692820	0.007276	0.824581
5	0.05	1	0.692820	0.001719	0.777316
5	0.1	1	0.692820	0.001238	0.776967
5	0.2	1	0.692820	0.000908	0.776269
5	0.4	1	0.692820	0.000687	0.774875

Success requires the maximum recorded 50 Hz feedback-envelope ratio and the synchronized exact-deadline error ratio both below 1-0.001, with no recorded tendon tension below 0.05 N. Raw values above one remain in the underlying records. All runs continued for 10 s. Jq uses the left-held applied command rate over the 500 actual 20-ms intervals; the t=10 s command evaluation has zero duration. It is a squared command-rate cost, not energy.

These are task-requirement evaluations, not independent statistical replicates or a proof of a continuous feasible region. B1 is independent of T and epsilon; its repeated trajectories coincide. Numerical state feedback preserves the original one-physics-step body-pose cache age. Exact-deadline and dense offline evaluations use separately synchronized physical states.

Display-table index: SD01. Full-precision source records are available under the data-availability arrangement on p. D1.

Complete nominal grid: B1

Data Table D2. All 28 task requirements for B1

T (s)	epsilon (m^-1)	Success (1/0)	Xmax control	Exact-T error/epsilon	10^3 Jq (m^2/s)
0.75	0.05	0	21.734935	21.850230	0.743485
0.75	0.1	0	10.923754	10.925115	0.743485
0.75	0.2	0	5.533151	5.462557	0.743485
0.75	0.4	0	2.784745	2.731279	0.743485
1	0.05	0	15.024217	15.013748	0.743485
1	0.1	0	7.512109	7.506874	0.743485
1	0.2	0	3.766492	3.753437	0.743485
1	0.4	0	1.910744	1.876718	0.743485
1.25	0.05	0	10.188127	10.118442	0.743485
1.25	0.1	0	5.142255	5.059221	0.743485
1.25	0.2	0	2.583347	2.529610	0.743485
1.25	0.4	0	1.312537	1.264805	0.743485
1.5	0.05	0	6.954782	6.950055	0.743485
1.5	0.1	0	3.495615	3.475027	0.743485
1.5	0.2	0	1.770615	1.737514	0.743485
1.5	0.4	1	0.904888	0.868757	0.743485
2	0.05	0	3.230119	3.222854	0.743485
2	0.1	0	1.638778	1.611427	0.743485
2	0.2	1	0.833564	0.805714	0.743485
2	0.4	1	0.692820	0.402857	0.743485
3	0.05	1	0.716790	0.704236	0.743485
3	0.1	1	0.692820	0.352118	0.743485
3	0.2	1	0.692820	0.176059	0.743485
3	0.4	1	0.692820	0.088030	0.743485
5	0.05	1	0.692820	0.038124	0.743485
5	0.1	1	0.692820	0.019062	0.743485
5	0.2	1	0.692820	0.009531	0.743485
5	0.4	1	0.692820	0.004765	0.743485

Success requires the maximum recorded 50 Hz feedback-envelope ratio and the synchronized exact-deadline error ratio both below 1-0.001, with no recorded tendon tension below 0.05 N. Raw values above one remain in the underlying records. All runs continued for 10 s. Jq uses the left-held applied command rate over the 500 actual 20-ms intervals; the t=10 s command evaluation has zero duration. It is a squared command-rate cost, not energy.

These are task-requirement evaluations, not independent statistical replicates or a proof of a continuous feasible region. B1 is independent of T and epsilon; its repeated trajectories coincide. Numerical state feedback preserves the original one-physics-step body-pose cache age. Exact-deadline and dense offline evaluations use separately synchronized physical states.

Display-table index: SD01. Full-precision source records are available under the data-availability arrangement on p. D1.

Complete nominal grid: B2

Data Table D3. All 28 task requirements for B2

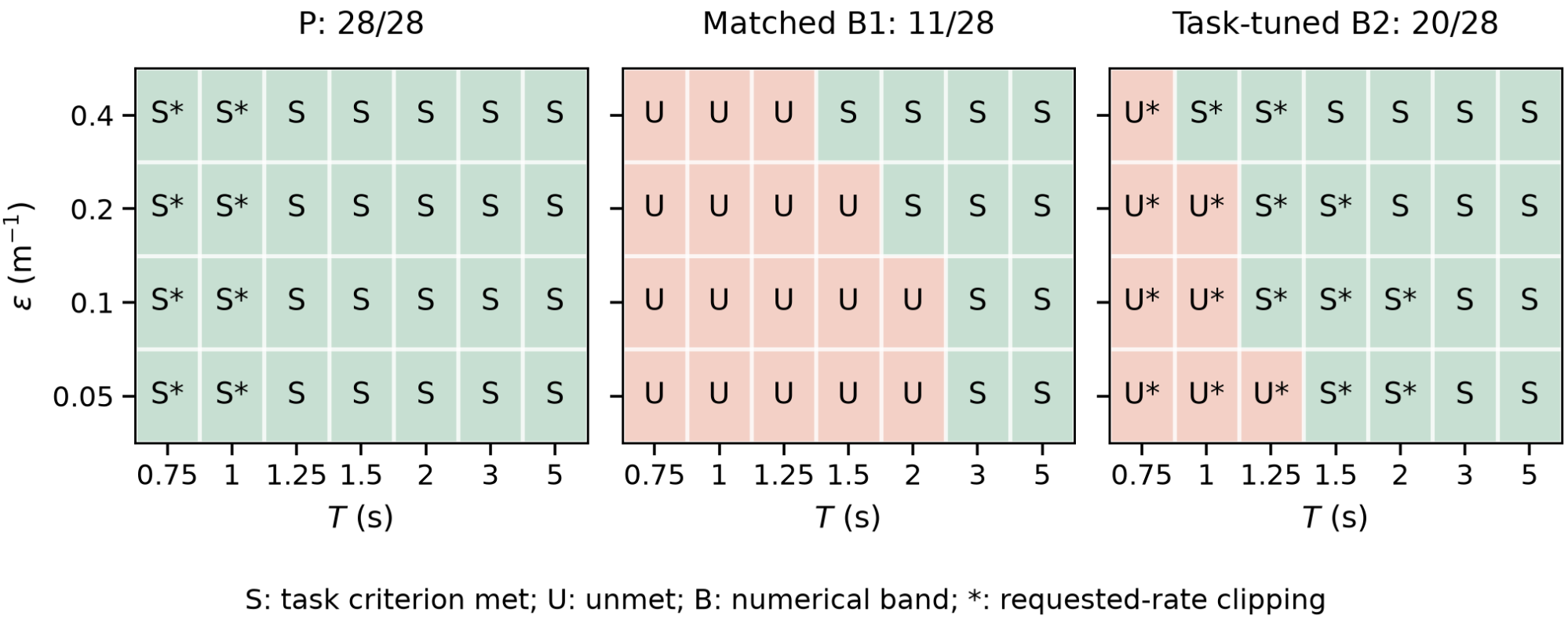
T (s)	epsilon (m^-1)	Success (1/0)	Xmax control	Exact-T error/epsilon	10^3 Jq (m^2/s)
0.75	0.05	0	2.828177	2.673397	1.365216
0.75	0.1	0	1.999786	1.876990	1.324505
0.75	0.2	0	1.442260	1.360436	1.265337
0.75	0.4	0	1.113319	1.042627	1.168603
1	0.05	0	1.437599	1.434623	1.295244
1	0.1	0	1.198306	1.169538	1.241550
1	0.2	0	1.022034	0.985187	1.162011
1	0.4	1	0.915817	0.876002	1.032861
1.25	0.05	0	1.058599	1.013866	1.225128
1.25	0.1	1	0.944046	0.904738	1.157876
1.25	0.2	1	0.879540	0.835043	1.058758
1.25	0.4	1	0.861010	0.814645	0.897412
1.5	0.05	1	0.881876	0.871909	1.155217
1.5	0.1	1	0.841017	0.818654	1.074586
1.5	0.2	1	0.824919	0.796600	0.955510
1.5	0.4	1	0.856366	0.820281	0.761953
2	0.05	1	0.763169	0.750283	1.015630
2	0.1	1	0.771617	0.752955	0.907754
2	0.2	1	0.812011	0.784466	0.749956
2	0.4	1	0.692820	0.402857	0.743485
3	0.05	1	0.716790	0.704236	0.743485
3	0.1	1	0.692820	0.352118	0.743485
3	0.2	1	0.692820	0.176059	0.743485
3	0.4	1	0.692820	0.088030	0.743485
5	0.05	1	0.692820	0.038124	0.743485
5	0.1	1	0.692820	0.019062	0.743485
5	0.2	1	0.692820	0.009531	0.743485
5	0.4	1	0.692820	0.004765	0.743485

Success requires the maximum recorded 50 Hz feedback-envelope ratio and the synchronized exact-deadline error ratio both below 1-0.001, with no recorded tendon tension below 0.05 N. Raw values above one remain in the underlying records. All runs continued for 10 s. Jq uses the left-held applied command rate over the 500 actual 20-ms intervals; the t=10 s command evaluation has zero duration. It is a squared command-rate cost, not energy.

These are task-requirement evaluations, not independent statistical replicates or a proof of a continuous feasible region. B1 is independent of T and epsilon; its repeated trajectories coincide. Numerical state feedback preserves the original one-physics-step body-pose cache age. Exact-deadline and dense offline evaluations use separately synchronized physical states.

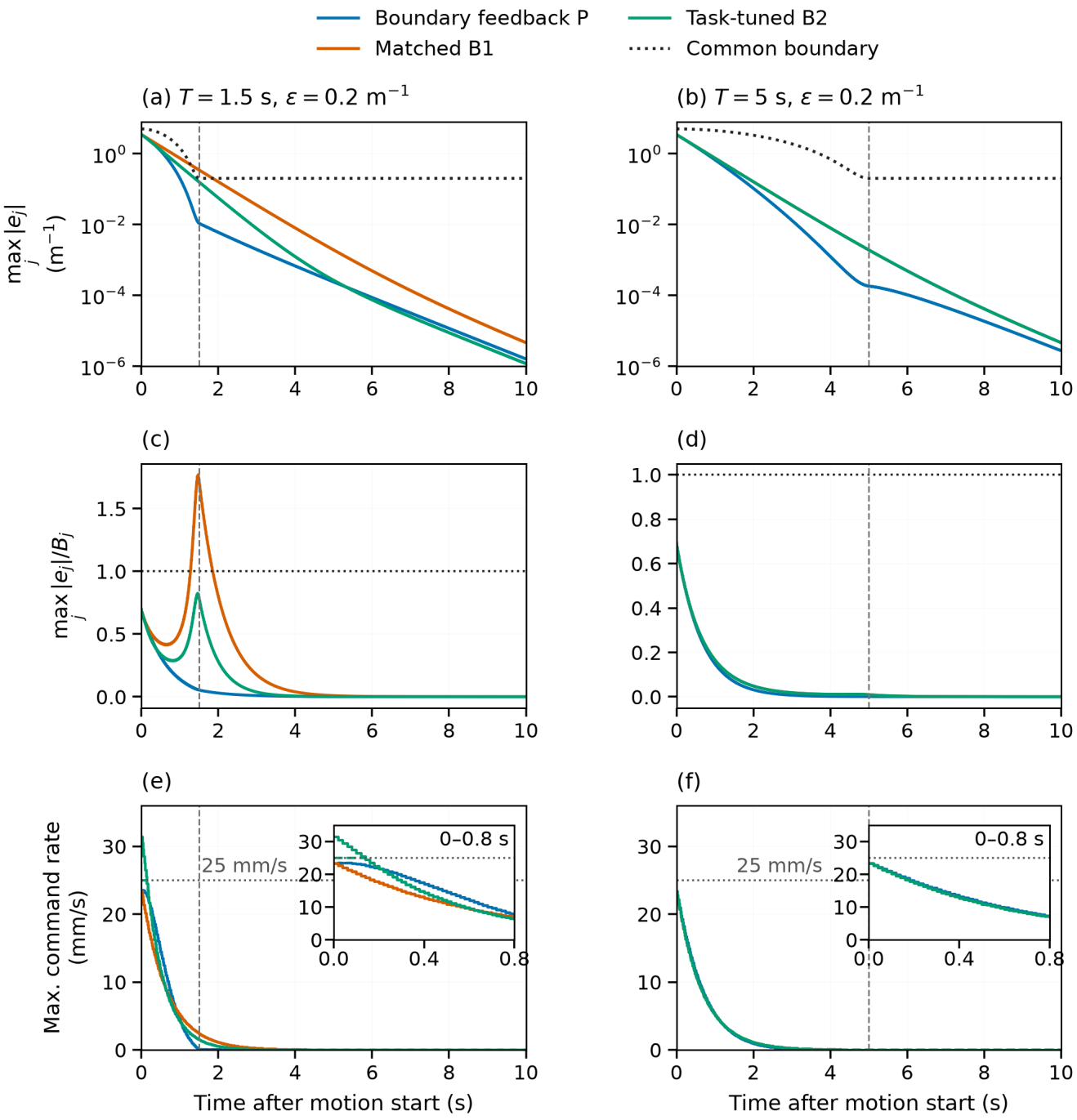
Display-table index: SD01. Full-precision source records are available under the data-availability arrangement on p. D1.

Data Figure D1. Complete nominal task grid



S marks a satisfied original sampled task, U an unmet task, B a numerical-band case, and an asterisk at least one requested-rate clip. The full 28-key grid is retained for every method. Counts are P 28/28, B1 11/28 and B2 20/28. A satisfied constrained run does not by itself verify the nominal theorem's accurate-execution assumption. All request, applied-command and force records remain available.

Data Figure D2. Prespecified nominal responses



Solid rates: requested; dashed rates: applied; vertical dashed lines: prescribed T

Upper panels: synchronized 2-ms CC error and common boundary (black dotted). Middle panels: synchronized normalized error and unit limit. Lower panels: requested (solid) and applied (dashed) maximum tendon-command rates, with the 25 mm/s limit. Blue, orange and green identify P, B1 and B2. Vertical dashed lines mark T. B1 and B2 coincide at T=5 s because their selected gains are equal. Insets show the initial 0.8 s; full 10 s windows remain visible. Logarithmic error axes end at 1e-6 m^-1; all underlying values remain in the source records.

Reproduction, integration and representative checks

Data Table D4. Representative joint half-step checks

Task/profile and method	Xmax dt=2 ms	Xmax dt=1 ms	Entry (s) dt=2 ms	Entry (s) dt=1 ms	Class unchanged
B1 T=0.75 e=0.05	21.85421	21.88941	2.762	2.763	Yes
B1 T=1.5 e=0.2	1.76941	1.77100	1.858	1.859	Yes
B1 T=5 e=0.2	0.69282	0.69282	1.858	1.859	Yes
B2 T=0.75 e=0.05	2.81971	2.83300	0.902	0.904	Yes
B2 T=1.5 e=0.2	0.82417	0.82495	1.386	1.386	Yes
B2 T=5 e=0.2	0.69282	0.69282	1.858	1.859	Yes
P T=0.75 e=0.05	0.69282	0.69282	0.678	0.681	Yes
P T=1.5 e=0.2	0.69282	0.69282	1.002	1.003	Yes
P T=5 e=0.2	0.69282	0.69282	1.662	1.663	Yes
Quartic T1.5 e=0.2	0.83497	0.83519	0.752	0.754	Yes
Quartic T5 e=0.2	0.69282	0.69282	1.202	1.204	Yes

All 11 independently repeated original-step records matched the corresponding numeric control samples exactly. These repeats verify determinism and are not extra statistical trials. The control period stayed 0.020 s when the physical step was halved. Representative dense validation covered 33 representative records. The complete nominal-grid evaluation of all 84 nominal records is reported in Data Table D22. Integration step and cached-feedback age change together; continuous-metric differences are provided in Data Table D26.

P0: unchanged archived-code reproduction

Run	Entry old / reproduced (s)	Largest numeric CSV difference
e1_fixed_main_30deg	1.68 / 1.68	1.77636e-15
e3_straight_crossing	0 / 0	1.77636e-15
e4_proposed	1.68 / 1.68	1.77636e-15
e4_baseline	1.86 / 1.86	1.77636e-15

P1: ideal RK4 error against the explicit solution

Step (s)	Maximum normalized discrepancy	Cases per step
0.002	1.401e-08	84
0.001	8.7104e-10	84
0.0005	5.4276e-11	84

Display-table index: SD02. Full-precision source records are available under the data-availability arrangement on p. D1.

Boundary-profile ablation

Data Table D5. Prespecified cubic/quartic comparison

T (s)	Profile	Entry control (s)	Entry dense (s)	Initial request (mm/s)	10^3 Jq (m^2/s)	Rate-clip time (s)
1.5	cubic	1.020	1.002	23.3827	1.030085	0.000
1.5	quartic	0.760	0.752	63.2891	1.312320	0.420
5	cubic	1.680	1.662	23.3827	0.776269	0.000
5	quartic	1.220	1.202	35.3546	1.036916	0.200

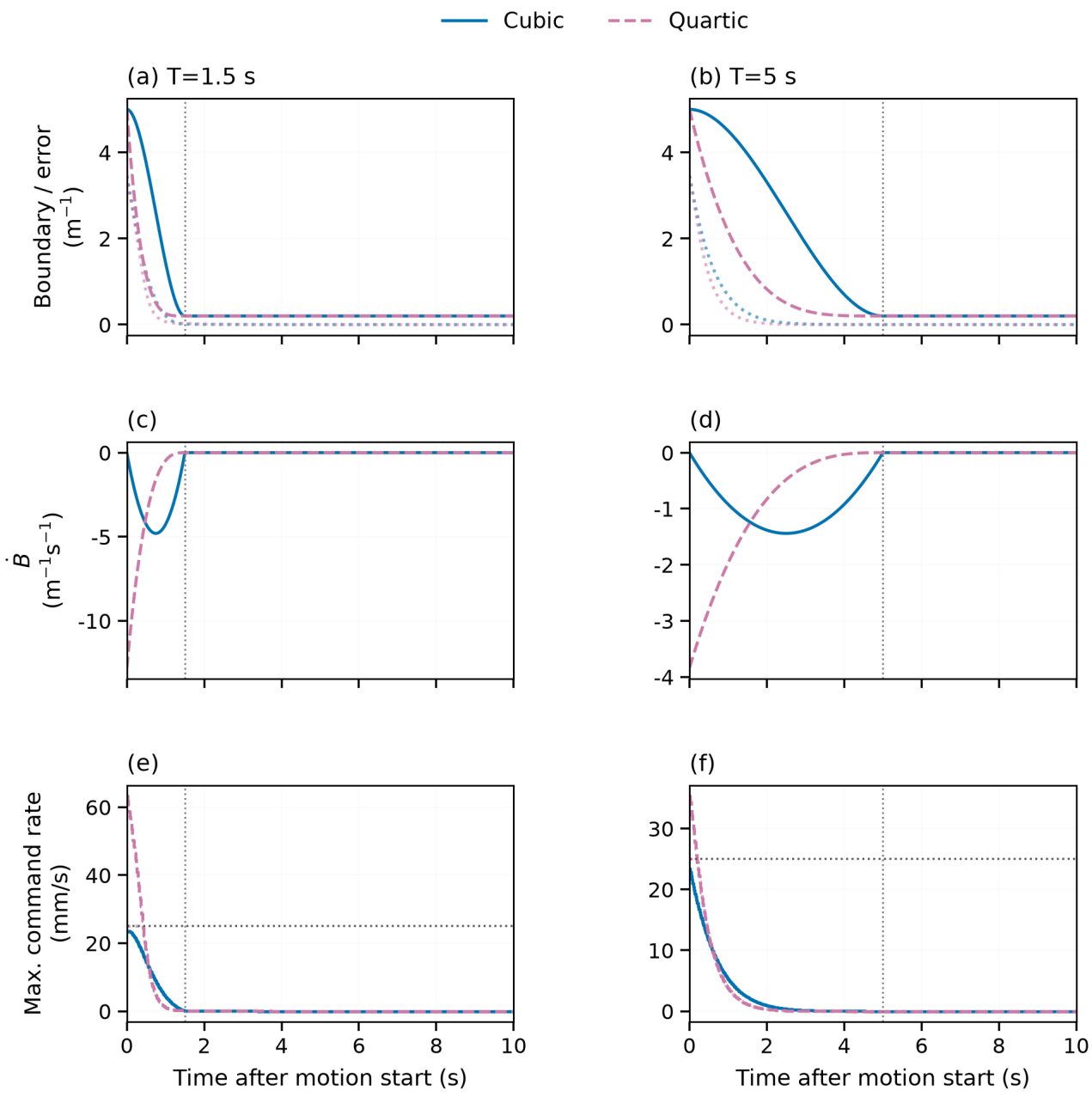
Both tasks use epsilon=0.2 m^-1, B0=5 m^-1, k=1.5 s^-1 and the same plant, initial state, estimator and input limits. Initial request is the largest absolute tendon rate at t=0 before clipping. Entry is the first strict terminal-band entry retained through the remaining record. Dense entry uses synchronized shape fits at every 0.002 s physical state. Rate-clip duration uses the recorded control flags and actual interval weights over the 10 s run.

The quartic profile is epsilon+(B0-epsilon)(1-t/T)^4 before T and constant afterwards. Its coefficients were checked against Zhu and Li (2022), Section 2.2, equations (11)-(12), DOI 10.3389/fams.2022.1041588. This comparison substitutes the boundary profile only; it does not reproduce their fuzzy learning controller.

The cubic boundary starts and ends with zero first derivative. The quartic boundary starts with a nonzero contraction rate and has a smoother higher-order terminal join. Its intermediate envelope differs from the cubic one. In these tasks the quartic profile enters earlier but has a larger initial request and higher command-rate cost. Independent repeats and physical-step halving preserved the reported classifications. The data support a tradeoff, not a general ranking of boundary profiles.

Display-table index: SD03. Full-precision source records are available under the data-availability arrangement on p. D1.

Data Figure D3. Boundary shape, contraction rate and command requests



Strong solid blue and dashed magenta lines identify the cubic and quartic profiles. Upper panels show each boundary (strong lines) and absolute feedback error (faint dotted lines). Middle panels show the boundary derivative. Lower panels show requested rates (strong lines), applied rates (faint dotted) and the 25 mm/s limit (black dotted). Vertical dotted lines mark T. Each profile has its own intermediate constraint and the same deadline and terminal width.

Paired joint-perturbation summary

Data Table D6. Ten fixed seeds per task and controller

T (s)	Method	n	Success n	Slack flags n	Max Xmax	Max deadline ratio	Mean 10^3 Jq	SD 10^3 Jq
1.5	P	10	3	2	2.99834	1.62699	1.037576	0.063709
1.5	B1	10	0	2	3.40841	2.61703	0.749598	0.056975
1.5	B2	10	4	2	3.09342	2.79599	0.969138	0.052287
5	P	10	8	2	0.69308	0.19605	0.788155	0.058933
5	B1	10	8	2	0.69308	0.47418	0.749598	0.056975
5	B2	10	8	2	0.69308	0.47418	0.749598	0.056975

Each task has epsilon=0.2 m^-1. The two tasks share ten prespecified seeds, 110601-110610. Parameters and pre-generated time-indexed node noise are paired across P, B1 and B2. The recovered original E5 mechanism samples stiffness, damping, actuator gain, pretension and controller radius factors uniformly within the archived ranges, with the archived noise-amplitude, delay and gravity selections. Separate deterministic parameter/noise random streams are documented in the machine-readable manifest.

Success uses the same combined feedback-envelope, synchronized exact-deadline and no-slack-flag criterion as the nominal grid. The slack flag is tension below 0.05 N at any saved synchronized physical state; it is not an extra calibrated sensor threshold. Costs are run-level squared command-rate integrals over [0,10] s. SD is the sample standard deviation over ten run-level values (ddof=1).

At T=1.5 s, P meets the task in 3/10 runs and task-tuned B2 in 4/10. At T=5 s all methods meet it in 8/10, with two seeds triggering tension flags. The complete unsuccessful cases are retained. These ten paired samples do not establish general robustness or statistical superiority.

Display-table index: SD04. Full-precision source records are available under the data-availability arrangement on p. D1.

Paired seed detail: T=1.5 s

Data Table D7. Complete per-seed results

Seed	Method	Success (1/0)	Xmax control	Deadline ratio	Entry control (s)	10^3 Jq	Clip (s)	Slack (s)
110601	P	0	2.77775	1.17186	3.780	1.052447	0.000	0.000
110601	B1	0	2.97540	1.72166	3.860	0.759854	0.000	0.000
110601	B2	0	2.56260	2.05809	3.400	0.999022	0.160	0.000
110602	P	0	1.96546	0.75888	4.100	1.068786	0.000	0.000
110602	B1	0	2.01073	0.87569	4.160	0.763106	0.000	0.000
110602	B2	0	1.70554	1.30079	3.600	0.994840	0.180	0.000
110603	P	0	2.84642	1.40856	3.300	1.105607	0.100	0.000
110603	B1	0	3.38460	2.61703	3.420	0.815153	0.020	0.000
110603	B2	0	2.94511	2.66953	3.020	1.040793	0.180	0.000
110604	P	1	0.69284	0.01710	0.960	1.119051	0.120	0.000
110604	B1	0	1.43919	1.36246	1.720	0.820135	0.000	0.000
110604	B2	1	0.69284	0.56981	1.280	1.020801	0.200	0.000
110605	P	0	1.12457	0.50126	3.040	0.994878	0.000	0.000
110605	B1	0	1.10075	0.22754	3.060	0.715384	0.000	0.000
110605	B2	1	0.98738	0.73432	0.960	0.929082	0.160	0.000
110606	P	0	0.87268	0.48531	0.820	0.904386	0.000	0.032
110606	B1	0	0.83785	0.28558	1.160	0.639743	0.000	0.010
110606	B2	0	0.85713	0.75582	0.940	0.858599	0.140	0.026
110607	P	1	0.69282	0.03865	0.980	1.067881	0.000	0.000
110607	B1	0	1.59639	1.56434	1.780	0.779207	0.000	0.000
110607	B2	1	0.72688	0.69858	1.340	0.983840	0.160	0.000
110608	P	0	2.99834	1.62699	3.460	0.978153	0.000	0.050
110608	B1	0	3.40841	2.52542	3.580	0.677755	0.000	0.032
110608	B2	0	3.09342	2.79599	3.120	0.935138	0.120	0.040
110609	P	0	1.04792	0.49759	2.540	1.039875	0.160	0.000
110609	B1	0	1.09578	0.59638	2.660	0.766585	0.040	0.000
110609	B2	0	1.00243	0.89697	1.800	0.962185	0.200	0.000
110610	P	1	0.69283	0.05446	1.000	1.044701	0.000	0.000
110610	B1	0	1.68805	1.65274	1.840	0.759057	0.000	0.000
110610	B2	1	0.78307	0.75657	1.380	0.967081	0.160	0.000

Every row is one complete 10 s execution. Entry uses the recorded control-feedback samples, whereas deadline ratio uses an independent fit at exactly T. A successful deadline value does not erase an earlier envelope violation or a tension flag. Costs use m^2/s before multiplication by 10^3 for display. Clip and slack durations are weighted by their actual logged intervals.

Display-table index: SD05. Full-precision source records are available under the data-availability arrangement on p. D1.

Paired seed detail: T=5 s

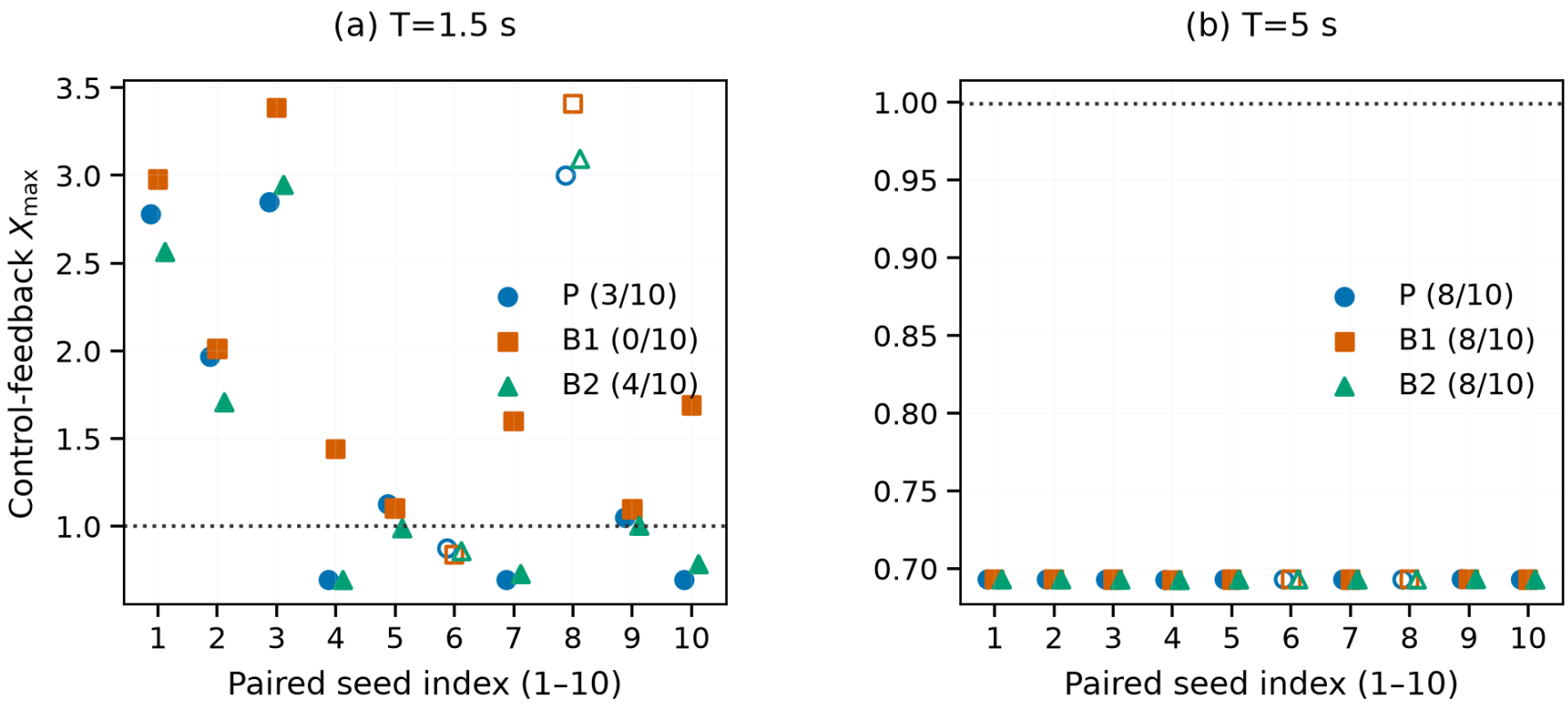
Data Table D7. Complete per-seed results (continued)

Seed	Method	Success (1/0)	Xmax control	Deadline ratio	Entry control (s)	10^3 Jq	Clip (s)	Slack (s)
110601	P	1	0.69289	0.04199	3.340	0.808449	0.000	0.000
110601	B1	1	0.69289	0.23824	3.860	0.759854	0.000	0.000
110601	B2	1	0.69289	0.23824	3.860	0.759854	0.000	0.000
110602	P	1	0.69287	0.16318	3.460	0.806293	0.000	0.000
110602	B1	1	0.69287	0.47418	4.160	0.763106	0.000	0.000
110602	B2	1	0.69287	0.47418	4.160	0.763106	0.000	0.000
110603	P	1	0.69274	0.00771	3.020	0.864973	0.020	0.000
110603	B1	1	0.69274	0.09141	3.420	0.815153	0.020	0.000
110603	B2	1	0.69274	0.09141	3.420	0.815153	0.020	0.000
110604	P	1	0.69252	0.00155	1.540	0.854225	0.020	0.000
110604	B1	1	0.69252	0.00402	1.720	0.820135	0.000	0.000
110604	B2	1	0.69252	0.00402	1.720	0.820135	0.000	0.000
110605	P	1	0.69285	0.19605	1.140	0.748688	0.000	0.000
110605	B1	1	0.69285	0.40399	3.060	0.715384	0.000	0.000
110605	B2	1	0.69285	0.40399	3.060	0.715384	0.000	0.000
110606	P	0	0.69291	0.06664	1.120	0.669211	0.000	0.012
110606	B1	0	0.69291	0.16390	1.160	0.639743	0.000	0.010
110606	B2	0	0.69291	0.16390	1.160	0.639743	0.000	0.010
110607	P	1	0.69282	0.00022	1.600	0.812153	0.000	0.000
110607	B1	1	0.69282	0.00467	1.780	0.779207	0.000	0.000
110607	B2	1	0.69282	0.00467	1.780	0.779207	0.000	0.000
110608	P	0	0.69285	0.01920	3.120	0.726296	0.000	0.034
110608	B1	0	0.69285	0.16860	3.580	0.677755	0.000	0.032
110608	B2	0	0.69285	0.16860	3.580	0.677755	0.000	0.032
110609	P	1	0.69308	0.11613	1.040	0.799549	0.060	0.000
110609	B1	1	0.69308	0.26037	2.660	0.766585	0.040	0.000
110609	B2	1	0.69308	0.26037	2.660	0.766585	0.040	0.000
110610	P	1	0.69283	0.00071	1.660	0.791717	0.000	0.000
110610	B1	1	0.69283	0.00811	1.840	0.759057	0.000	0.000
110610	B2	1	0.69283	0.00811	1.840	0.759057	0.000	0.000

Every row is one complete 10 s execution. Entry uses the recorded control-feedback samples, whereas deadline ratio uses an independent fit at exactly T. A successful deadline value does not erase an earlier envelope violation or a tension flag. Costs use m^2/s before multiplication by 10^3 for display. Clip and slack durations are weighted by their actual logged intervals.

Display-table index: SD05. Full-precision source records are available under the data-availability arrangement on p. D1.

Data Figure D4. All paired perturbation outcomes



Points show control-sample Xmax. Hollow symbols mark recorded tension below 0.05 N. Legend counts additionally require deadline accuracy and no tension flag. The right panel magnifies the region near the initial-error maximum. P, B1 and B2 share the same seed-level parameters and noise at each task. All sixty outcomes, including unfavorable and mixed comparisons, are retained.

Hardware: formal trials and calibration

Data Table D8. Six formal trials, original 2 Hz feedback layer

Trial	Entry (s)	Terminal RMSE (m^-1)	Max |xi|	Final bias (m^-1)	Return residual (m^-1)	Command cost
P1	7.0006	0.032633	0.982096	0.000276	0.049869	7.676608
B1	6.0038	0.046633	0.771949	-0.008462	0.086224	6.615766
P2	6.5020	0.025653	0.718684	0.014454	0.055812	7.154870
B2	7.5017	0.036108	0.970399	-0.008930	0.071438	7.520860
P3	6.5007	0.021956	0.710269	-0.010001	0.092302	6.702223
B3	6.5032	0.036076	0.920114	0.024395	0.045151	7.600261

Each trial has 24 recorded control updates, a 12 s tracking event and T=8 s. Terminal RMSE uses the recorded control samples at t>=8 s; final bias uses the final 1 s control window. Return residual uses the final 1 s return-vision mean relative to the frozen software zero. Command cost uses motor-command coordinate differences and real intervals (command-degree squared per second), not measured motor energy or physical angle.

Controller-level descriptive statistics (three trials each)

Method	n	RMSE mean (m^-1)	RMSE sample SD	Entry mean (s)	Entry sample SD
baseline	3	0.0396060	0.0060858	6.669572	0.762666
proposed	3	0.0267471	0.0054221	6.667760	0.288257

The average per-trial terminal RMSE reduction is 32.466866%. This is a descriptive observation with n=3, not a frame-level sample-size claim.

Data Table D9. Six calibration responses

Calibration ID	Command (deg coordinate)	Baseline n	Steady n	Upper response (m^-1)
CAL10_R1	10	89	30	-0.5813281
CAL15_R1	15	87	29	-0.8508334
CAL10_R2	10	78	30	-0.5761715
CAL15_R2	15	90	30	-0.8629053
CAL10_R3	10	87	28	-0.5887262
CAL15_R3	15	86	29	-0.8686990

Only command levels 10 and 15 were independently tested, three times each. The fitted gain is -0.05574745603 m^-1 per command-degree, intercept -0.02460073819 m^-1 and R-squared 0.997895256. The formal target corresponds to 7.901020 command-degrees, outside those two levels. No independent nearby static validation group was found. The high R-squared does not independently validate that extrapolation.

Display-table index: SD06. Full-precision source records are available under the data-availability arrangement on p. D1.

Hardware: signal levels, timing and boundary audit

Data Table D10. Both recorded observation levels

Trial	Signal layer	n	Max |xi|	Exceedances	Minimum estimated error margin (m^-1)
P1	2 Hz median	24	0.982096	0	0.0041086
P1	EMA-marker frame fit	317	1.044606	4	-0.0074929
B1	2 Hz median	24	0.771949	0	0.0772990
B1	EMA-marker frame fit	258	0.960888	0	0.0122715
P2	2 Hz median	24	0.718684	0	0.0643670
P2	EMA-marker frame fit	360	1.088700	5	-0.0190809
B2	2 Hz median	24	0.970399	0	0.0058488
B2	EMA-marker frame fit	342	1.408716	34	-0.0764733
P3	2 Hz median	24	0.710269	0	0.0773719
P3	EMA-marker frame fit	354	1.102048	13	-0.0263396
B3	2 Hz median	24	0.920114	0	0.0188810
B3	EMA-marker frame fit	359	1.209483	24	-0.0516507

All 144 recorded recent-median control samples are inside the boundary. Of 1,990 frame-level CC fits from EMA marker coordinates, 80 exceed it in five of the six trials. Those frame-level fits are a distinct signal layer; neither is a direct unfiltered truth measurement. The two CSV columns called raw and filtered curvature are identical and both derive from the EMA-marker fit.

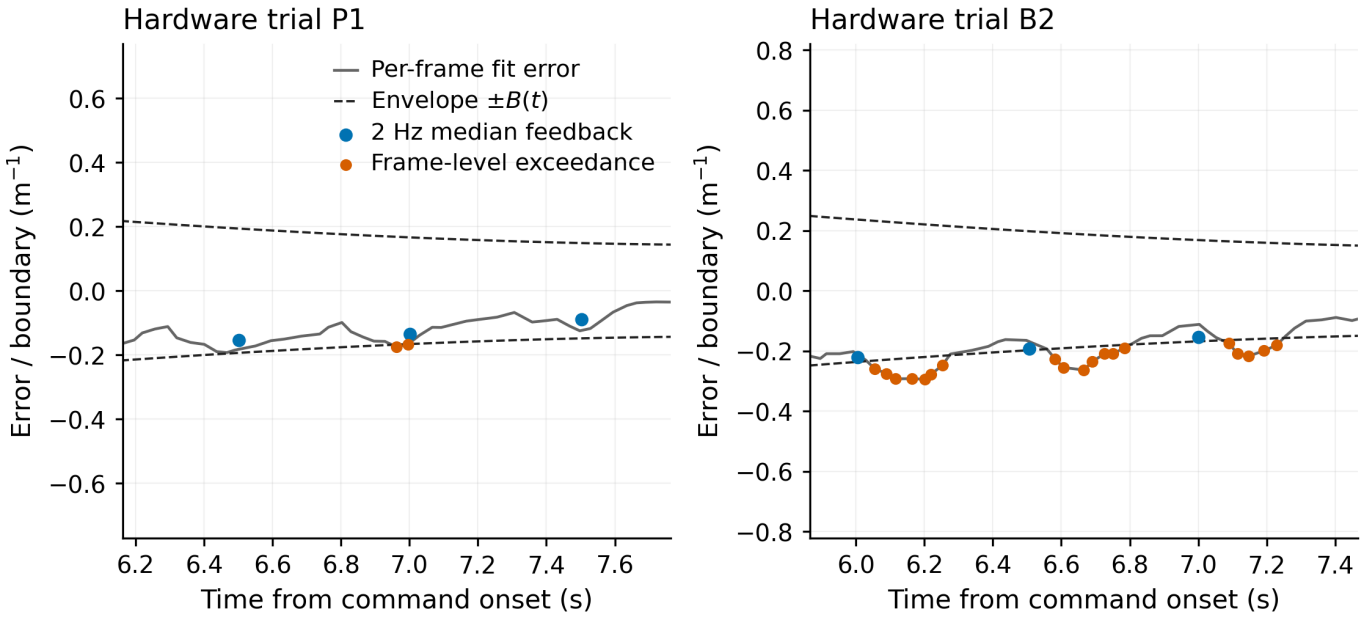
Data Table D11. Actual tracking/control intervals

Trial	Valid frame samples	Frame median / max interval (s)	Control median / max interval (s)	Minimum estimated control error margin (m^-1)
P1	317	0.03618 / 0.08973	0.50005 / 0.50318	0.0041086
B1	258	0.03701 / 0.14177	0.50000 / 0.50552	0.0772990
P2	360	0.03460 / 0.04884	0.50054 / 0.50871	0.0643670
B2	342	0.03076 / 0.07212	0.50085 / 0.50904	0.0058488
P3	354	0.03446 / 0.07690	0.49958 / 0.51472	0.0773719
B3	359	0.03544 / 0.04915	0.50011 / 0.50433	0.0188810

The recorded chain is marker detection, coordinate EMA (0.42 new sample + 0.58 previous state), CC fitting, median of the latest 0.2 s valid curvature window, and nominal 2 Hz control. The median window is anchored to the latest vision timestamp. Exposure and processing-completion timestamps were not separately saved, so these intervals are not an end-to-end latency bound. The largest static H0 sample SD is 0.0355617 m^-1; it is not a deterministic dynamic error bound and exceeds the P1 control minimum margin. No continuous-time true-curvature compliance conclusion follows.

Display-table index: SD07. Full-precision source records are available under the data-availability arrangement on p. D1.

Data Figure D5. Hardware trial P1 and B2: frame-level peaks



Frame-level CC fits and control-feedback samples are shown over the critical tracking intervals. Both occur during tracking and before the return phase. P1 reaches |xi|=1.044606 at t=6.962562 s; hardware trial B2 reaches |xi|=1.408716 at t=6.665613 s. The original 2 Hz recent-median results remain visible as a different observation level. Here B2 denotes the second hardware baseline trial.

Hardware: clipping, protocol adjustment and warning

Data Table D12. Per-trial update counts (denominator 24)

Trial	Rate clip	Step clip	Position clip	Quantized	Protocol collision
P1	16	0	0	9	1
B1	12	0	0	13	4
P2	15	0	0	10	2
B2	15	0	0	10	4
P3	14	0	0	11	1
B3	15	0	0	10	2

Data Table D13. Time-weighted command-layer fractions

Trial	Rate clip (%)	Step clip (%)	Position clip (%)	Quantization (%)	Protocol collision (%)
P1	66.6751	0.0000	0.0000	37.5050	4.1801
B1	50.0320	0.0000	0.0000	54.1319	16.6561
P2	62.5328	0.0000	0.0000	41.6586	8.3142
B2	62.5142	0.0000	0.0000	41.6184	16.5878
P3	58.3348	0.0000	0.0000	45.8022	4.1370
B3	62.5160	0.0000	0.0000	41.6271	8.2869

Counts use 24 actual control updates per trial. Time weights hold each logged applied target until the next update, with the last target held to the documented 12 s endpoint. This is the command-target time base, not a continuous motor-position or velocity measurement. Several restrictions can overlap on the same update and their counts must not be added as unique limited steps.

Requested, controller-applied and protocol-applied targets were recorded. Separate rate, step and protocol flags were reconstructed offline only because the frozen algebraic controller, state and selector were available. Reconstruction agrees with all 144 applied values to within 1.776e-15 and reproduces the saved frame strings. The reconstruction is explicitly distinguished from an originally recorded flag stream.

Retained P3 return warning

The runtime recent 0.5 s median return residual was -0.1136714 m^-1, beyond its 3-sigma threshold of 0.1066851 m^-1. The offline final 1 s mean residual was 0.0923016 m^-1 and met that different-window criterion. Both results are retained; the runtime warning is not erased by the offline statistic. Physical motor-speed units, tension and continuous encoder trajectories were not established by these historical logs.

Display-table index: SD08. Full-precision source records are available under the data-availability arrangement on p. D1.

Hardware: complete recorded attempt ledger

Data Table D14. All 27 recorded archive entries

Archive time 6 August 2026	Stage	Method	Disposition	Recorded reason
Preliminary	Preliminary	--	Outside formal	No unified run metadata
10:37:15	Static	--	Safety abort	Operator interrupt
10:39:19	Static	--	Outside formal	Separate stage
10:41:14	Static	--	Outside formal	Separate stage
10:42:32	Static	--	Outside formal	Separate stage
11:09:37	Static	--	Outside formal	Separate stage
11:10:55	Static	--	Outside formal	Separate stage
11:12:11	Static	--	Outside formal	Separate stage
13:30:57	Preflight	--	Outside formal	Separate stage
13:47:57	Calibration 10deg	Open loop	Outside formal	Separate stage
13:53:11	Calibration 15deg	Open loop	Outside formal	Separate stage
13:55:22	Calibration 10deg	Open loop	Outside formal	Separate stage
13:57:55	Calibration 15deg	Open loop	Outside formal	Separate stage
14:01:44	Calibration 10deg	Open loop	Outside formal	Separate stage
14:05:25	Calibration 15deg	Open loop	Outside formal	Separate stage
14:09:50	pilot	P	Safety abort	Motion-time exception
14:27:43	pilot	P	Safety abort	Motion-time GT1 timeout
15:16:36	pilot	P	Outside formal	Separate stage
15:21:06	pilot	B	Outside formal	Separate stage
15:24:23	formal	P	Accepted	Complete formal record
15:29:16	formal	B	Safety abort	Marker/CC loss >0.3 s
15:44:04	formal	B	Accepted	Complete formal record
15:47:58	formal	P	Technical exclusion	Low processing FPS; complete record retained
15:57:53	formal	P	Accepted	Complete formal record
16:01:08	formal	B	Accepted	Complete formal record
16:03:52	formal	P	Accepted	Complete formal record
16:07:50	formal	B	Accepted	Complete formal record

The compact ID is the archive timestamp; the full attempt_id and all provenance fields remain in the companion H2 CSV. The located scope is 26 metadata-bearing run folders plus one preliminary-material folder. It comprises 7 static records, 1 preflight, 6 calibrations, 4 pilots and 8 formal-stage attempts. Six formal trials were accepted; one formal marker-loss abort sent no tracking target, and one complete formal attempt was excluded for processing rate. Safety aborts are retained separately. This is not a claim that all historical attempts anywhere are known. Documented criteria are not described as preregistered.

Display-table index: SD09. Full-precision source records are available under the data-availability arrangement on p. D1.

Discretization audit: historical snapshot and unresolved convergence

Data Table D15. Recomputed original 3 s snapshots

N	Shape RMSE (mm)	Tip difference to N24 (mm)	Curvature-vector difference (m^-1)	Original numerical thresholds met
8	0.3467954	1.7065755	0.0236987	No
12	0.3654357	1.1236634	0.0378840	Yes
16	0.1250337	1.0587858	0.0213854	Yes
24	0.6263297	0.0000000	0.0000000	Yes

Shape RMSE is the RMS Euclidean node-to-fitted-CC distance within each model, using n_fit=N+1 internal/base nodes, not a difference from N24 and not a denominator of 3*n_fit. Tip and curvature differences use N24 as the reference. N12 also meets the old 1.2 mm/0.1 m^-1 thresholds; N16 was not their unique passing model. The original N24 reference is step-sensitive, so this table remains a historical snapshot, not a validated convergence limit.

Data Table D16. Common 20 s fixed-command recheck

N	Physics step (ms)	Shape RMSE (mm)	Tip difference to N24 (mm)	Curvature difference (m^-1)	Endpoint velocity norm (rad/s)
8	1	0.3438903	0.696516	0.0200063	1.3596e-11
8	0.5	0.3438903	0.696516	0.0200063	1.5152e-11
12	1	0.7378611	15.516468	0.4028816	0.0026693
12	0.5	1.1342067	20.870866	0.5863720	0.0020487
16	1	0.0925477	0.182171	0.0032193	1.5325e-07
16	0.5	0.0925532	0.182153	0.0032198	9.865e-08
24	1	0.0442642	0.000000	0.0000000	8.3063e-13
24	0.5	0.0442642	0.000000	0.0000000	4.3641e-13

Data Table D17. Cross-step and final-hold sensitivity

N	Halving tip difference (mm)	Halving curvature difference (m^-1)	19 to 20 s tip drift at 1 ms (mm)	19 to 20 s tip drift at 0.5 ms (mm)
8	4.508355e-10	6.958188e-10	5.379236e-09	6.169442e-09
12	6.240455	0.2087036	0.5778618	0.5593893
16	5.439785e-05	1.704287e-06	4.456216e-05	1.157769e-05
24	1.532852e-13	3.836258e-11	9.714451e-14	2.254874e-13

The common re-evaluation used fixed model and actuator settings, a 2-s pretension stage, a 20-s command hold and physics steps of 1 and 0.5 ms. The N = 12 model retained cross-step and final-hold differences. Tables D15-D17 report the original fixed-time comparison and the longer-hold checks separately.

Display-table index: SD10. Full-precision source records are available under the data-availability arrangement on p. D1.

Numerical metrics and the original 56-run composition

Data Table D18. E3 tip-error windows

Window	Samples	Tip RMS (mm)
t < 5 s	250	1.129011402
All [0,12] s	601	1.305035431
t >= 5 s	351	1.417138602

All windows use the same norm of p(desired curvature)-p_MuJoCo. The 250 and 351 samples do not overlap at the boundary. The weighted squared-RMS identity closes to 4.44e-16 mm^2. The separate whole-record delta_CC RMS is 0.187249590 mm and is not substituted for tip tracking error. A larger later-window RMS is allowed for this time-varying task.

Data Table D19. Full-record saturation denominator (radius +10%)

Definition	Fraction	Percent
Mean over 501 times x 6 actuator flags	0.002661343979	0.2661344
Any tendon clipped at a recorded time	0.003992015968	0.3992016

The old saturation value 0.0027 is a rounded full-record actuator-sample fraction, not a terminal-window or any-tendon fraction. Endpoint counting also inflated old slack durations: the three affected joint runs 06/11/12 change from 10.02, 10.00, 10.02 s to 10.00, 9.98, 10.00 s under observed-interval weights. These remain sampled diagnostics, not continuous exact slack observation.

Data Table D20. E4 cost from unrounded command-rate records

Method	Trapezoid Jq (m^2/s)	Command-difference Jq (m^2/s)
proposed	0.0007762685996227	0.0007762685996228
baseline	0.0007434846115232	0.0007434846115232

The unrounded trapezoid values give a cost increase of 4.409504594%. The alternative command-difference integral agrees to numerical precision. The 4.41% statement is supported by unrounded records, not by reverse calculation from a six-decimal display. Jq is not an energy measure.

Data Table D21. Original 56 numerical runs

Category	Count
E0_static_mapping	12
E1_fixed_target	1
E1_six_direction_repeat	6
E2_time_varying_tracking	1
E3_straight_crossing	1
E4_controller_comparison	2
E5_robustness_joint_random	20
E5_robustness_oat	13

The non-E5 categories sum to 23; E5 comprises 13 one-at-a-time plus 20 joint-random runs. Total: 56. E6 and all 168 comparison executions, as well as four P0 reproductions, are separate. Full run_id-level provenance is retained in the source ledger.

Display-table index: SD11. Full-precision source records are available under the data-availability arrangement on p. D1.

Complete nominal criterion check: P

Data Table D22. Full 84-record correspondence

T (s)	epsilon (m^-1)	Xmax control	Xmax dense	Exact-T ratio	Control gate/strict	Dense gate/strict	Raw dense exceedance	Xmax band control/dense
0.75	0.05	0.69282	0.69282	0.10651	1/1	1/1	0.00000	below/below
0.75	0.1	0.69282	0.69282	0.16800	1/1	1/1	0.00000	below/below
0.75	0.2	0.69282	0.69282	0.21121	1/1	1/1	0.00000	below/below
0.75	0.4	0.69282	0.69282	0.23157	1/1	1/1	0.00000	below/below
1	0.05	0.69282	0.69282	0.06854	1/1	1/1	0.00000	below/below
1	0.1	0.69282	0.69282	0.08990	1/1	1/1	0.00000	below/below
1	0.2	0.69282	0.69282	0.10747	1/1	1/1	0.00000	below/below
1	0.4	0.69282	0.69282	0.12119	1/1	1/1	0.00000	below/below
1.25	0.05	0.69282	0.69282	0.05405	1/1	1/1	0.00000	below/below
1.25	0.1	0.69282	0.69282	0.06504	1/1	1/1	0.00000	below/below
1.25	0.2	0.69282	0.69282	0.07438	1/1	1/1	0.00000	below/below
1.25	0.4	0.69282	0.69282	0.08223	1/1	1/1	0.00000	below/below
1.5	0.05	0.69282	0.69282	0.04350	1/1	1/1	0.00000	below/below
1.5	0.1	0.69282	0.69282	0.04904	1/1	1/1	0.00000	below/below
1.5	0.2	0.69282	0.69282	0.05391	1/1	1/1	0.00000	below/below
1.5	0.4	0.69282	0.69282	0.05816	1/1	1/1	0.00000	below/below
2	0.05	0.69282	0.69282	0.02604	1/1	1/1	0.00000	below/below
2	0.1	0.69282	0.69282	0.02672	1/1	1/1	0.00000	below/below
2	0.2	0.69282	0.69282	0.02757	1/1	1/1	0.00000	below/below
2	0.4	0.69282	0.69282	0.02853	1/1	1/1	0.00000	below/below
3	0.05	0.69282	0.69282	0.00970	1/1	1/1	0.00000	below/below
3	0.1	0.69282	0.69282	0.00851	1/1	1/1	0.00000	below/below
3	0.2	0.69282	0.69282	0.00774	1/1	1/1	0.00000	below/below
3	0.4	0.69282	0.69282	0.00728	1/1	1/1	0.00000	below/below
5	0.05	0.69282	0.69282	0.00172	1/1	1/1	0.00000	below/below
5	0.1	0.69282	0.69282	0.00124	1/1	1/1	0.00000	below/below
5	0.2	0.69282	0.69282	0.00091	1/1	1/1	0.00000	below/below
5	0.4	0.69282	0.69282	0.00069	1/1	1/1	0.00000	below/below

Each row is one nominal execution, identified by E7_method_Tvalue_epsvalue. Control uses 501 recorded feedback samples; dense uses all 5001 synchronized 2-ms node states. The exact-T column is the independent recorded synchronized fit. Gate means both relevant ratios are below 0.999 and no synchronized tension is below 0.05 N; strict replaces both ratio thresholds by 1. Each gate/strict cell lists those two binary decisions. Dense uses its sequential synchronized CC fit, including its own exact-T value.

The numerical band is [0.999,1.001]; below/above refer to Xmax relative to that band. Raw exceedance is max(0,Xmax_dense-1), independently of the gate. No run falls in the numerical band. Across all methods, original, strict and dense decisions agree task by task: P28/28, B1 11/28 and B2 20/28. No dense fit failed.

Display-table index: SD12. Full-precision source records are available under the data-availability arrangement on p. D1.

Complete nominal criterion check: B1

Data Table D22. Full 84-record correspondence (continued)

T (s)	epsilon (m^-1)	Xmax control	Xmax dense	Exact-T ratio	Control gate/strict	Dense gate/strict	Raw dense exceedance	Xmax band control/dense
0.75	0.05	21.73493	21.85421	21.85023	0/0	0/0	20.85421	above/above
0.75	0.1	10.92375	10.93873	10.92511	0/0	0/0	9.93873	above/above
0.75	0.2	5.53315	5.52925	5.46256	0/0	0/0	4.52925	above/above
0.75	0.4	2.78474	2.78278	2.73128	0/0	0/0	1.78278	above/above
1	0.05	15.02422	15.01375	15.01375	0/0	0/0	14.01375	above/above
1	0.1	7.51211	7.50770	7.50687	0/0	0/0	6.50770	above/above
1	0.2	3.76649	3.76387	3.75344	0/0	0/0	2.76387	above/above
1	0.4	1.91074	1.90941	1.87672	0/0	0/0	0.90941	above/above
1.25	0.05	10.18813	10.18111	10.11844	0/0	0/0	9.18111	above/above
1.25	0.1	5.14225	5.13871	5.05922	0/0	0/0	4.13871	above/above
1.25	0.2	2.58335	2.58157	2.52961	0/0	0/0	1.58157	above/above
1.25	0.4	1.31254	1.31163	1.26481	0/0	0/0	0.31163	above/above
1.5	0.05	6.95478	6.95112	6.95005	0/0	0/0	5.95112	above/above
1.5	0.1	3.49561	3.49324	3.47503	0/0	0/0	2.49324	above/above
1.5	0.2	1.77062	1.76941	1.73751	0/0	0/0	0.76941	above/above
1.5	0.4	0.90489	0.90427	0.86876	1/1	1/1	0.00000	below/below
2	0.05	3.23012	3.22799	3.22285	0/0	0/0	2.22799	above/above
2	0.1	1.63878	1.63770	1.61143	0/0	0/0	0.63770	above/above
2	0.2	0.83356	0.83301	0.80571	1/1	1/1	0.00000	below/below
2	0.4	0.69282	0.69282	0.40286	1/1	1/1	0.00000	below/below
3	0.05	0.71679	0.71636	0.70424	1/1	1/1	0.00000	below/below
3	0.1	0.69282	0.69282	0.35212	1/1	1/1	0.00000	below/below
3	0.2	0.69282	0.69282	0.17606	1/1	1/1	0.00000	below/below
3	0.4	0.69282	0.69282	0.08803	1/1	1/1	0.00000	below/below
5	0.05	0.69282	0.69282	0.03812	1/1	1/1	0.00000	below/below
5	0.1	0.69282	0.69282	0.01906	1/1	1/1	0.00000	below/below
5	0.2	0.69282	0.69282	0.00953	1/1	1/1	0.00000	below/below
5	0.4	0.69282	0.69282	0.00477	1/1	1/1	0.00000	below/below

Each row is one nominal execution, identified by E7_method_Tvalue_epsvalue. Control uses 501 recorded feedback samples; dense uses all 5001 synchronized 2-ms node states. The exact-T column is the independent recorded synchronized fit. Gate means both relevant ratios are below 0.999 and no synchronized tension is below 0.05 N; strict replaces both ratio thresholds by 1. Each gate/strict cell lists those two binary decisions. Dense uses its sequential synchronized CC fit, including its own exact-T value.

The numerical band is [0.999,1.001]; below/above refer to Xmax relative to that band. Raw exceedance is max(0,Xmax_dense-1), independently of the gate. No run falls in the numerical band. Across all methods, original, strict and dense decisions agree task by task: P28/28, B1 11/28 and B2 20/28. No dense fit failed.

Display-table index: SD12. Full-precision source records are available under the data-availability arrangement on p. D1.

Complete nominal criterion check: B2

Data Table D22. Full 84-record correspondence (continued)

T (s)	epsilon (m^-1)	Xmax control	Xmax dense	Exact-T ratio	Control gate/strict	Dense gate/strict	Raw dense exceedance	Xmax band control/dense
0.75	0.05	2.82818	2.81971	2.67340	0/0	0/0	1.81971	above/above
0.75	0.1	1.99979	1.99486	1.87699	0/0	0/0	0.99486	above/above
0.75	0.2	1.44226	1.43944	1.36044	0/0	0/0	0.43944	above/above
0.75	0.4	1.11332	1.11169	1.04263	0/0	0/0	0.11169	above/above
1	0.05	1.43760	1.43590	1.43462	0/0	0/0	0.43590	above/above
1	0.1	1.19831	1.19621	1.16954	0/0	0/0	0.19621	above/above
1	0.2	1.02203	1.02060	0.98519	0/0	0/0	0.02060	above/above
1	0.4	0.91582	0.91484	0.87600	1/1	1/1	0.00000	below/below
1.25	0.05	1.05860	1.05694	1.01387	0/0	0/0	0.05694	above/above
1.25	0.1	0.94405	0.94278	0.90474	1/1	1/1	0.00000	below/below
1.25	0.2	0.87954	0.87857	0.83504	1/1	1/1	0.00000	below/below
1.25	0.4	0.86101	0.86028	0.81464	1/1	1/1	0.00000	below/below
1.5	0.05	0.88188	0.88077	0.87191	1/1	1/1	0.00000	below/below
1.5	0.1	0.84102	0.84010	0.81865	1/1	1/1	0.00000	below/below
1.5	0.2	0.82492	0.82417	0.79660	1/1	1/1	0.00000	below/below
1.5	0.4	0.85637	0.85577	0.82028	1/1	1/1	0.00000	below/below
2	0.05	0.76317	0.76248	0.75028	1/1	1/1	0.00000	below/below
2	0.1	0.77162	0.77100	0.75295	1/1	1/1	0.00000	below/below
2	0.2	0.81201	0.81147	0.78447	1/1	1/1	0.00000	below/below
2	0.4	0.69282	0.69282	0.40286	1/1	1/1	0.00000	below/below
3	0.05	0.71679	0.71636	0.70424	1/1	1/1	0.00000	below/below
3	0.1	0.69282	0.69282	0.35212	1/1	1/1	0.00000	below/below
3	0.2	0.69282	0.69282	0.17606	1/1	1/1	0.00000	below/below
3	0.4	0.69282	0.69282	0.08803	1/1	1/1	0.00000	below/below
5	0.05	0.69282	0.69282	0.03812	1/1	1/1	0.00000	below/below
5	0.1	0.69282	0.69282	0.01906	1/1	1/1	0.00000	below/below
5	0.2	0.69282	0.69282	0.00953	1/1	1/1	0.00000	below/below
5	0.4	0.69282	0.69282	0.00477	1/1	1/1	0.00000	below/below

Each row is one nominal execution, identified by E7_method_Tvalue_epsvalue. Control uses 501 recorded feedback samples; dense uses all 5001 synchronized 2-ms node states. The exact-T column is the independent recorded synchronized fit. Gate means both relevant ratios are below 0.999 and no synchronized tension is below 0.05 N; strict replaces both ratio thresholds by 1. Each gate/strict cell lists those two binary decisions. Dense uses its sequential synchronized CC fit, including its own exact-T value.

The numerical band is [0.999,1.001]; below/above refer to Xmax relative to that band. Raw exceedance is max(0,Xmax_dense-1), independently of the gate. No run falls in the numerical band. Across all methods, original, strict and dense decisions agree task by task: P28/28, B1 11/28 and B2 20/28. No dense fit failed.

Display-table index: SD12. Full-precision source records are available under the data-availability arrangement on p. D1.

Perturbation failure categories

Data Table D23. Run-level category counts (denominator ten per row)

T (s)	Method	n	Met	Envelope fail	Deadline fail	Post-T outside	Tension flag	Estimator fail	Numerical fail
1.5	B1	10	0	9	6	9	2	0	0
1.5	B2	10	4	5	4	5	2	0	0
1.5	P	10	3	6	3	6	2	0	0
5	B1	10	8	0	0	0	2	0	0
5	B2	10	8	0	0	0	2	0	0
5	P	10	8	0	0	0	2	0	0

Failure categories overlap. Envelope failure is control-feedback Xmax>=0.999; deadline failure is synchronized exact-T ratio>=0.999; the post-T column counts a strict terminal-band exit at any recorded control sample on [T,10] s. Tension flags use any synchronized physical-state value below 0.05 N. Estimator failure is an unsuccessful recorded control or deadline fit; numerical failure is non-completion or a non-finite physical record. Original and strict combined-task counts coincide in all sixty executions.

At T=1.5 s, the proposed and task-tuned methods satisfy 3/10 and 4/10 cases, respectively. At T=5 s, all three satisfy 8/10, with two seeds triggering the tension threshold. These descriptive counts retain the original task and seeds; overlapping categories cannot be summed into a unique failure count.

A separate temporary-entry diagnostic records any exit after the first terminal-band entry anywhere in the record. Such an exit may occur before T and can therefore coexist with a satisfied task. It is kept separately from post-T failure in Data Table D24 and the full CSV.

Display-table index: SD13. Full-precision source records are available under the data-availability arrangement on p. D1.

Per-seed failure decomposition: T=1.5 s

Data Table D24. All 60 seed-level reason flags

Seed	Method	Met	Envelope fail	Deadline fail	Post-T outside	Temporary entry/exit	Tension flag	Fit fail	Num. fail
110601	P	0	1	1	1	1	0	0	0
110601	B1	0	1	1	1	1	0	0	0
110601	B2	0	1	1	1	1	0	0	0
110602	P	0	1	0	1	1	0	0	0
110602	B1	0	1	0	1	1	0	0	0
110602	B2	0	1	1	1	1	0	0	0
110603	P	0	1	1	1	1	0	0	0
110603	B1	0	1	1	1	1	0	0	0
110603	B2	0	1	1	1	1	0	0	0
110604	P	1	0	0	0	0	0	0	0
110604	B1	0	1	1	1	0	0	0	0
110604	B2	1	0	0	0	0	0	0	0
110605	P	0	1	0	1	1	0	0	0
110605	B1	0	1	0	1	1	0	0	0
110605	B2	1	0	0	0	0	0	0	0
110606	P	0	0	0	0	0	1	0	0
110606	B1	0	0	0	0	0	1	0	0
110606	B2	0	0	0	0	0	1	0	0
110607	P	1	0	0	0	0	0	0	0
110607	B1	0	1	1	1	0	0	0	0
110607	B2	1	0	0	0	0	0	0	0
110608	P	0	1	1	1	1	1	0	0
110608	B1	0	1	1	1	1	1	0	0
110608	B2	0	1	1	1	1	1	0	0
110609	P	0	1	0	1	1	0	0	0
110609	B1	0	1	0	1	1	0	0	0
110609	B2	0	1	0	1	1	0	0	0
110610	P	1	0	0	0	0	0	0	0
110610	B1	0	1	1	1	0	0	0	0
110610	B2	1	0	0	0	0	0	0	0

One means the stated outcome occurred, zero means it did not. Met is the original combined task gate; all remaining columns are diagnostic flags, with overlap. A temporary entry/exit before T is distinct from the post-T terminal-band condition. Fit denotes a recorded estimator failure; Num. denotes numerical non-completion or non-finite states. Error maxima, deadline ratios, cost and durations for these same sixty runs appear in Data Table D7 and in the full-precision current CSV.

Display-table index: SD14. Full-precision source records are available under the data-availability arrangement on p. D1.

Per-seed failure decomposition: T=5 s

Data Table D24. All 60 seed-level reason flags (continued)

Seed	Method	Met	Envelope fail	Deadline fail	Post-T outside	Temporary entry/exit	Tension flag	Fit fail	Num. fail
110601	P	1	0	0	0	1	0	0	0
110601	B1	1	0	0	0	1	0	0	0
110601	B2	1	0	0	0	1	0	0	0
110602	P	1	0	0	0	1	0	0	0
110602	B1	1	0	0	0	1	0	0	0
110602	B2	1	0	0	0	1	0	0	0
110603	P	1	0	0	0	1	0	0	0
110603	B1	1	0	0	0	1	0	0	0
110603	B2	1	0	0	0	1	0	0	0
110604	P	1	0	0	0	0	0	0	0
110604	B1	1	0	0	0	0	0	0	0
110604	B2	1	0	0	0	0	0	0	0
110605	P	1	0	0	0	0	0	0	0
110605	B1	1	0	0	0	1	0	0	0
110605	B2	1	0	0	0	1	0	0	0
110606	P	0	0	0	0	0	1	0	0
110606	B1	0	0	0	0	0	1	0	0
110606	B2	0	0	0	0	0	1	0	0
110607	P	1	0	0	0	0	0	0	0
110607	B1	1	0	0	0	0	0	0	0
110607	B2	1	0	0	0	0	0	0	0
110608	P	0	0	0	0	1	1	0	0
110608	B1	0	0	0	0	1	1	0	0
110608	B2	0	0	0	0	1	1	0	0
110609	P	1	0	0	0	0	0	0	0
110609	B1	1	0	0	0	1	0	0	0
110609	B2	1	0	0	0	1	0	0	0
110610	P	1	0	0	0	0	0	0	0
110610	B1	1	0	0	0	0	0	0	0
110610	B2	1	0	0	0	0	0	0	0

One means the stated outcome occurred, zero means it did not. Met is the original combined task gate; all remaining columns are diagnostic flags, with overlap. A temporary entry/exit before T is distinct from the post-T terminal-band condition. Fit denotes a recorded estimator failure; Num. denotes numerical non-completion or non-finite states. Error maxima, deadline ratios, cost and durations for these same sixty runs appear in Data Table D7 and in the full-precision current CSV.

Display-table index: SD14. Full-precision source records are available under the data-availability arrangement on p. D1.

Original E5 and paired perturbation definitions

Data Table D25. Definition and observation-level comparison

Definition	Original E5	Paired perturbation P4
Record unit	13 single-factor plus 20 joint-parameter runs; P only	10 paired seeds x 2 deadlines x 3 methods
Task	T=5 s; epsilon=0.2 m^-1; target 4[cos(30 deg), sin(30 deg)] m^-1; 10 s record	T=1.5 or 5 s; same epsilon/target/window
Parameters and seeds	Joint seeds 20260760–20260779; original sequential random streams	Seeds 110601–110610; same archived ranges; per-seed matched parameters and external time-indexed node-noise arrays
Envelope statement	violation_count=sum(max_j|xi_j|>=1) on 50 Hz feedback; all 33 recorded maxima below 1	Raw envelope exceedances retained; original combined gate requires Xmax<0.999
Deadline	Control-grid suffix entry recorded; no independent synchronized exact-T gate	Independent physical t=T CC fit; ratio<0.999 in original gate
Tension	Reported separately on control samples; <0.05 N in 3 old runs; not part of zero-envelope-violation statement	Any <0.05 N at 2-ms synchronized states fails combined criterion
Endpoint duration	Old durations counted sample flags times 0.02 s, including final endpoint	Actual adjacent-interval weights; t=10 endpoint has zero duration
Inference	Finite task/seed set descriptive curvature result	Finite paired task/seed set descriptive combined outcome; not a like-for-like robustness rate

The original E5 curvature statement used recorded 50-Hz feedback-envelope violations and kept tension as a separate metric. Its 33 records all have feedback maxima below 1, while three contain a control-sample tension flag. P4 combines envelope, synchronized exact-deadline accuracy and dense tension in a different paired task/seed set. The two reported outcomes therefore have different denominators and definitions.

Display-table index: SD15. Full-precision source records are available under the data-availability arrangement on p. D1.

N16 joint integration and feedback-age sensitivity

Data Table D26. Eleven original/half-step pairs; ordered entries are 2 ms / 1 ms

Task/profile and method	Entry (s) 2 ms / 1 ms	Dense Xmax 2 ms / 1 ms	10^3 Jq (m^2/s) 2 ms / 1 ms	Jq change (%)	Max tip diff (mm)	RMS tip diff (mm)	Tip diff at T (mm)
B1 T=0.75 eps=0.05	2.762 / 2.763	21.85421 / 21.88941	0.743485 / 0.743314	-0.022968	1.520633	0.093094	0.047701
B1 T=1.5 eps=0.2	1.858 / 1.859	1.76941 / 1.77100	0.743485 / 0.743314	-0.022968	1.520633	0.093094	0.012683
B1 T=5 eps=0.2	1.858 / 1.859	0.69282 / 0.69282	0.743485 / 0.743314	-0.022968	1.520633	0.093094	0.000919
B2 T=0.75 eps=0.05	0.902 / 0.904	2.81971 / 2.83300	1.365216 / 1.365334	0.008632	1.625783	0.116121	0.028912
B2 T=1.5 eps=0.2	1.386 / 1.386	0.82417 / 0.82495	0.955510 / 0.955438	-0.007504	1.625783	0.104707	0.003265
B2 T=5 eps=0.2	1.858 / 1.859	0.69282 / 0.69282	0.743485 / 0.743314	-0.022968	1.520633	0.093094	0.000919
P T=0.75 eps=0.05	0.678 / 0.681	0.69282 / 0.69282	1.459248 / 1.459415	0.011443	1.616980	0.117500	0.006393
P T=1.5 eps=0.2	1.002 / 1.003	0.69282 / 0.69282	1.030085 / 1.029949	-0.013177	1.520784	0.109650	0.004376
P T=5 eps=0.2	1.662 / 1.663	0.69282 / 0.69282	0.776269 / 0.776099	-0.021850	1.520633	0.095119	0.000495
Quartic T1.5 eps=0.2	0.752 / 0.754	0.83497 / 0.83519	1.312320 / 1.312433	0.008674	1.625783	0.115759	0.013534
Quartic T5 eps=0.2	1.202 / 1.204	0.69282 / 0.69282	1.036916 / 1.036900	-0.001551	1.625783	0.108404	0.001218

The control period remains 20 ms. Halving the physics step also changes the cached-feedback age from 2 to 1 ms, so these pairs measure joint implementation sensitivity. All eleven task classifications remain unchanged. Entry changes by at most 3.000 ms; the largest absolute relative Jq change is 0.022968%. The largest absolute dense-Xmax change is 0.035198.

Tip differences compare synchronized actual tip positions at identical common 2 ms times. RMS is computed over all 5001 common times. The maximum difference is 1.625783 mm; the largest pairwise RMS is 0.117500 mm. Peak differences occur in early transients; the largest deadline difference is 0.047701 mm. Per-pair peak times, initial differences and final 10 s differences remain in the CSV. Numerical sensitivity of continuous metrics is reported separately from binary classification agreement.

Display-table index: SD16. Full-precision source records are available under the data-availability arrangement on p. D1.